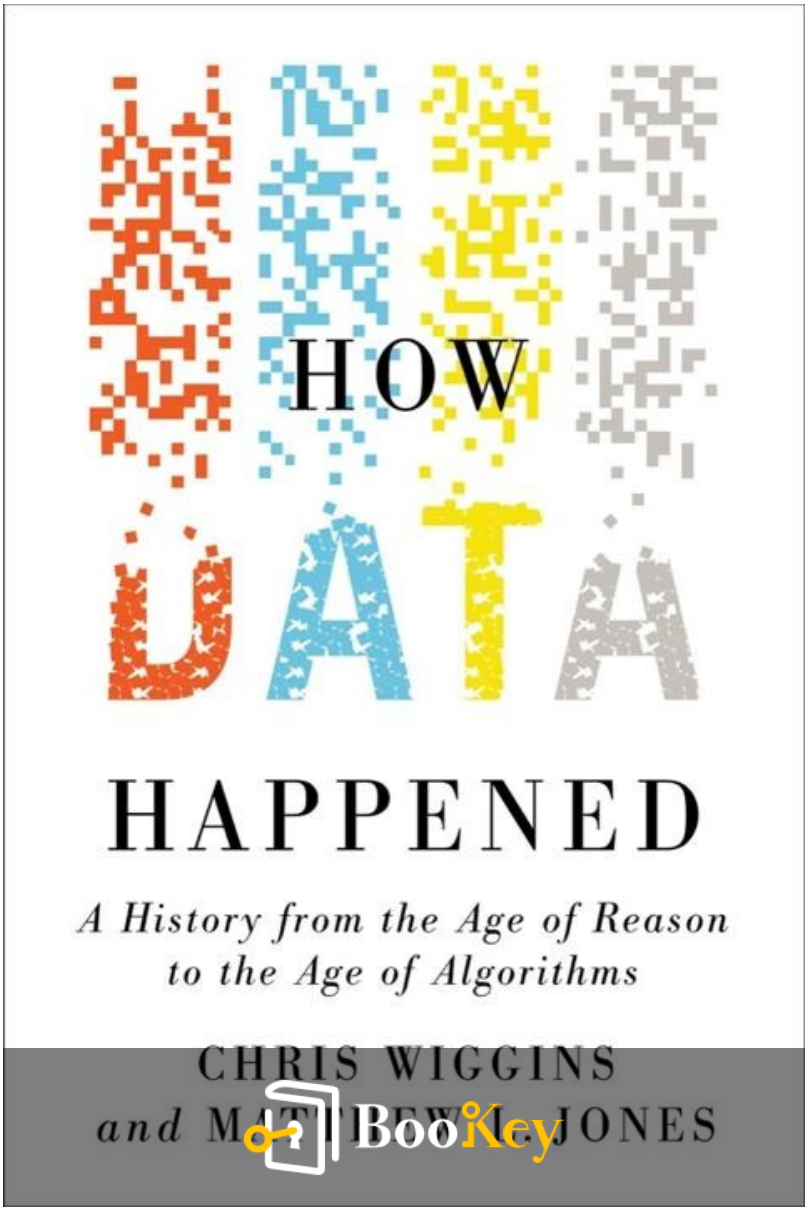
How Data Happened PDF

Chris Wiggins



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How Data Happened

Unraveling Data's Influence on Society, Power, and Ethics.

Written by Bookey

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About the book

In "How Data Happened," Chris Wiggins and Matthew L. Jones provide an expansive exploration of the history and impact of data, revealing its profound technical, political, and ethical implications in our society. From the use of facial recognition technology to automated decision-making systems that influence critical life outcomes, the authors trace the roots of our data-driven world back through centuries of history—from the foundational US census to the origins of eugenics and the rise of search engines like Google. Drawing from their acclaimed course at Columbia University, Wiggins and Jones examine how data has served both as a powerful tool for truth and a means of entrenching authority. They argue that by understanding the historical trajectory of data, we can intentionally shape its future to reflect our collective values and aspirations.

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About the author

Chris Wiggins is a distinguished data scientist and professor known for his impactful contributions to the fields of data science, mathematics, and physics. Serving as an associate professor at Columbia University, Wiggins combines his rich academic background with extensive experience in applying data-driven methodologies across various disciplines. His work is characterized by a commitment to making data comprehensible and accessible, promoting the importance of data literacy in an increasingly data-driven world. Additionally, Wiggins has been a key figure in initiatives that bridge academia and industry, emphasizing the ethical use of data and fostering a deeper understanding of its societal implications. His insights, garnered through years of research and teaching, enrich "How Data Happened," providing readers with a thoughtful exploration of the evolution and significance of data in contemporary life.

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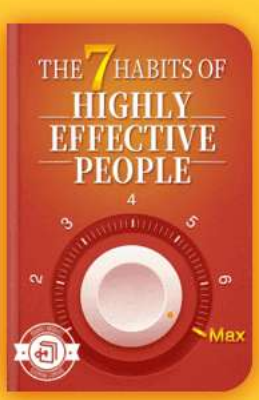
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Summary Content List

Chapter 1 : The Stakes

Chapter 2 : Social Physics and l’homme moyen

Chapter 3 : The Statistics of the Deviant

Chapter 4 : Data, Intelligence, and Policy

Chapter 5 : Data’s Mathematical Baptism

Chapter 6 : Data at War

Chapter 7 : Intelligence without Data

Chapter 8 : Volume, Variety, and Velocity

Chapter 9 : Machines, Learning

Chapter 10 : The Science of Data

Chapter 11 : The Battle for Data Ethics

Chapter 12 : Persuasion, Ads, and Venture Capital

Chapter 13 : Solutions beyond Solutionism

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Chapter 1 Summary : The Stakes



Section	Summary
Overview of Data's Impact	The chapter emphasizes the dual nature of technology, advocating for collaboration between technologists and social scientists to address algorithmic biases.
Historical Context	Critiques by figures like Safiya Noble and Cathy O'Neil focus on data biases and their impact on marginalized communities, highlighting ethical concerns related to corporate data practices and government surveillance.
Emergence of Data Science	Voices like Ruha Benjamin and Zeynep Tufekci stress the need for awareness in data practices, arguing that data often reinforces societal inequalities and challenges the idea of objectivity.
Kranzberg's Law of Technology	The author highlights the need to critically analyze technology's implications, citing historical figures who shaped statistical methods that affected power structures and governance.
Contemporary Implications	The chapter calls for scrutiny of data practices in terms of fairness and accountability, advocating for institutional responses and political action to combat systemic inequalities.
Conclusion	Chapter 1 sets the foundation for understanding the complex relationship between data, technology, and society, asserting data's role in shaping values and social structures.

Chapter 1: The Stakes

Overview of Data's Impact

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The chapter introduces the complexities surrounding data's role in society, emphasizing that technology is neither purely beneficial nor detrimental. Advocates for a data-centered revolution, like Hanna Wallach, call for collaboration between technologists and social scientists to address biases in algorithmic systems.

Historical Context

The chapter traces historical critiques of data's influence on social dynamics. Prominent figures such as Safiya Noble and Cathy O'Neil highlight the challenges posed by data biases and the consequences of algorithmic decisions on marginalized communities. The rise of corporate data practices, alongside government surveillance, raises ethical concerns about power dynamics.

Emergence of Data Science

Significant voices, including Ruha Benjamin and Zeynep Tufekci, stress the need for awareness and accountability in data practices. The chapter argues that the data-driven nature of modern technology often reinforces societal inequalities,

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challenging the notion of objectivity in data collection and analysis.

Kranzberg's Law of Technology

The author invokes Kranzberg’s first law, emphasizing the importance of critically analyzing technology's implications. Historical figures, including Quetelet, Galton, and Pearson, created foundational statistical approaches that both reinforced and challenged existing power structures. The chapter outlines how statistical rigor became intertwined with societal governance and policy formulation.

Contemporary Implications

Data practices are scrutinized through the lens of fairness and accountability, with increasing recognition of the systemic inequalities they perpetuate. The need for comprehensive institutional responses and political action is underscored, recognizing that the trajectory of data practices can be altered through conscious effort.
Overall, Chapter 1 sets the stage for understanding the intricate relationship between data, technology, and society, positioning data as a pivotal factor that shapes values, power, and social structures.

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Example

Key Point:Acknowledging Data's Dual Nature

Example:Imagine you are working on an important project, relying heavily on data to drive your conclusions. As you gather insights, you notice that some trends favor one demographic over another, highlighting the necessity of critically assessing your data sources and methodologies. This experience underscores how embracing data's complexities requires collaboration and ethical scrutiny to ensure equitable outcomes.

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Critical Thinking

Key Point:Data's duality in shaping social structures.

Critical Interpretation:The chapter highlights that data technology has a complex role in society, asserting that its impact is neither wholly positive nor negative. The emphasis on collaboration between technologists and social scientists suggests a nuanced understanding of data applications, which may also be interpreted as a call for responsibility in data science. However, critics may argue that the author's reliance on certain voices overlooks alternative perspectives that might emphasize the potentially transformative power of data without the same caution. For instance, proponents of data-driven innovation may contend that data can indeed lead to significant societal advancement, challenging the view that it primarily perpetuates inequalities (Noble, 2018; O'Neil, 2016). Therefore, while the argument for sensitivity towards data's impact is valid, it is crucial to consider the potential for positive change through responsible data use.

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Chapter 2 Summary : Social Physics and l’homme moyen



Chapter 2: Social Physics and l’homme moyen

Overview of Social Physics

- Adolphe Quetelet, a Belgian astronomer, established the study of social physics, emphasizing the importance of data in understanding societal phenomena.
- Influential figures like Florence Nightingale echoed Quetelet's belief in the need for data-driven reforms in governance, urging policymakers to consider the empirical effects of laws.

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The Importance of Numbers

- In 19th-century Europe, the emergence of statistical data was seen as a critical method to understand societal trends and human characteristics.
- Critics historically decried the use of numbers to define social conditions, arguing for a qualitative understanding of society.

Quetelet's Contributions

- Quetelet introduced the concept of the "average man" (l’homme moyen) and pushed for statistical methods to understand social phenomena.
- He advocated for the gradual improvement of societal conditions through reform based on statistical observations rather than revolutionary actions.

Transition to Modern Statistics

- As governments began to harness the power of statistical data, categories like race and class became increasingly tied to quantitative metrics, shifting how society understood

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itself.
- The emergence of statistical laws led to the reification of averages, causing societal constructs to be viewed as natural laws.

Influence of Francis Galton

- Quetelet's work influenced Francis Galton, who expanded upon eugenics through statistical methods, advocating for the improvement of the human race via data.
- Galton's applications of statistics laid the groundwork for future studies of intelligence based on measurable criteria.

Statistics in Governance

- By the early 20th century, governing bodies and institutions exploited statistical models to justify social policies and interventions, particularly concerning race and class.
- The use of statistics became entrenched in bureaucratic practices, leading to a reliance on quantitative measures in policy decisions.

Statistical Analysis and Social Impact

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- As statistical techniques evolved, they granted authorities the power to dictate social hierarchies, offering a veneer of legitimacy to prejudiced views.
- The ongoing tension between data interpretation and social justice concerns highlights the potential for statistics to both empower and oppress.

Conclusion

- The development of social physics and its intertwining with statistics transformed societal governance, creating a landscape where data-driven decisions became the norm.
- Understanding these historical patterns is vital for navigating contemporary issues regarding data ethics, representation, and accountability in an increasingly data-centric world.

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Chapter 3 Summary : The Statistics of the Deviant

Chapter 3: The Statistics of the Deviant

Overview of Data and Power

The chapter explores the evolution of statistical practices and their intersection with societal structures, specifically focusing on how statistics has been utilized to understand and control human behavior. It traces the historical roots of quantitative analysis, highlighting figures like Adolphe Quetelet, Francis Galton, and Karl Pearson, and illustrates how their work laid the foundation for modern statistical approaches.

Quetelet and Social Physics

Quetelet, a Belgian astronomer, introduced the concept of the "average man" and sought to apply statistical methods to societal observations. He advocated for a gradual

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transformation of society through data-driven insights rather than abrupt upheaval. His work established a foundation for viewing numerical data as a critical means of understanding social phenomena.

The Birth of Eugenics

The chapter discusses how Quetelet's ideas influenced Galton, who focused on hereditary differences and the development of eugenics. Galton asserted that intelligence and other attributes could be inherited, pushing for societal improvement through the selective breeding of “superior” individuals. His belief in eugenics propagated racist theories by asserting differences in abilities among races.

Udny Yule and Causal Correlation

Udny Yule applied advanced statistical methods to analyze

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Chapter 4 Summary : Data, Intelligence, and Policy

Section	Summary
Overview of Data and Algorithmic Systems	This section discusses the impact of data-driven technologies on governance, fairness, and accountability, highlighting concerns about data reinforcing social inequalities.
Emergence of Data Critique	Highlights figures like Wallach, Noble, and O'Neil who critique data practices, calling for fair and representative models that do not perpetuate stereotypes or ethical issues in decision-making.
Historical Context of Data Utilization	Examines the evolution of data collection from the 19th century to present, discussing its use in social engineering and eugenics.
Key Figures and Concepts	Notable figures include Quetelet (social physics), Galton & Pearson (eugenics via statistics), and Yule (causation vs correlation in public assistance).
Data Privacy and Government Surveillance	Analyzes privacy concerns post-Watergate and 9/11, discussing the balancing act between individual rights and corporate data practices.
Transformation of Data Science	Discusses shifts in data science with the integration of big data analytics and machine learning, led by figures like Tukey and Hammerbacher.
Industrial Data Science and Growing Challenges	Observes the evolution of data science as emphasizing decision-making over scientific rigor, warning against biases inherent in training data.
Conclusion	Cautions against viewing data and algorithms as neutral, advocating for ethical practices that consider the impact on marginalized communities amidst the tension of progress and fairness.

Chapter 4 Summary: Data, Intelligence, and Policy

Overview of Data and Algorithmic Systems

This chapter examines how data-driven technologies impact governance, fairness, and accountability in societal systems. Central figures in the dialogue about the intersection of data,

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technology, and policy include Hanna Wallach, Safiya Noble, Cathy O’Neil, and Ruha Benjamin, highlighting concerns that data collection can reinforce existing social inequalities.

Emergence of Data Critique

The chapter recounts Wallach's speech at a conference where she introduced ideas about integrating social science with machine learning to ensure that models are fair and representative of society. Noble criticized Google's search algorithms for perpetuating stereotypes particularly against marginalized groups. Meanwhile, O’Neil's warnings about the dangers of algorithms in decision-making emphasize the ethical implications of data science.

Historical Context of Data Utilization

The chapter analyzes the evolution of data collection, from governmental systems and census data to contemporary practices in commercial data collection. Beginning with the establishment of statistical methods in the 19th century, the chapter delves into how statistical practices morphed into tools for eugenics and social engineering.

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Key Figures and Concepts

-
Adolphe Quetelet
used mathematics to advocate for "social physics," focusing on the average individual. His work laid the groundwork for quantitative sociology.
-
Francis Galton
and
Karl Pearson
shaped eugenics and biological heredity through statistics, underlining the importance of data and correlation in understanding societal issues.
-
Udny Yule's
regression analysis posed significant questions about the relationships between public assistance and poverty, highlighting the ongoing methodological debates about causation versus correlation.

Data Privacy and Government Surveillance

The chapter highlights the historical backdrop of privacy

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concerns following events like Watergate and 9/11 that legitimized increasing surveillance and data collection practices by both corporations and governments. Ground-breaking legislative efforts like the Privacy Act of 1974 reflect a complex battle between safeguarding individual rights and enabling corporate data practices.

Transformation of Data Science

Amidst discussions about the challenges of bias and inequality, significant shifts in data science emerged by the early 21st century, as fields began to embrace big data analytics and machine learning techniques to process vast amounts of information and glean insights. Figures like John Tukey and later, Jeff Hammerbacher, contributed to the ethos that data-driven discovery could transform industries and academic disciplines.

Industrial Data Science and Growing Challenges

Finally, the chapter observes how the rise of "data science" as a profession evolved from traditional statistics, emphasizing decision-making over strict adherence to scientific rigor. Yet, it warns against complacency, stressing

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that the algorithms created—while powerful in predictive capabilities—remain subject to biases from the very data they are trained on, underscoring a need for an ongoing examination of ethical implications and the societal impact of these technologies.

Conclusion

The chapter cautions against the assumption that data and algorithms are neutral, advocating instead for a nuanced approach that prioritizes ethical data science practices and considers the tangible effects on society, particularly on marginalized communities. The tension between utilizing data for progress while ensuring fairness and accountability remains a central theme in contemporary discussions surrounding technology and policy.

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Example

Key Point:The assumption that data and algorithms are inherently neutral is misleading.

Example:Imagine you're applying for a job through an online platform that uses algorithms to filter candidates. You submit your application, excited about the opportunity, but unknown to you, the algorithm favors applicants from certain educational backgrounds due to historical hiring patterns. As a result, deserving candidates like you, who may not fit that mold, miss the chance simply because the underlying data used to train the algorithm reflects existing inequalities. This situation illustrates how data-driven decision-making isn't just about analytics; it's an ethical issue tied to fairness and societal impact, making it crucial to recognize the biases in data and strive for inclusive practices that account for diverse experiences.

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Critical Thinking

Key Point:The neutrality of data and algorithms is a misleading assumption that needs critical reevaluation.

Critical Interpretation:The chapter challenges the perception that data-driven technologies operate free from bias, advocating for closer scrutiny of their societal implications, particularly regarding marginalized communities. For instance, Safiya Noble's critique of algorithmic biases reveals how data can perpetuate stereotypes rather than promote fairness. This perspective invites readers to question the presumed objectivity of data, emphasizing the importance of ethical considerations in data science. Critics like Cathy O’Neil have also documented the consequences of biased algorithms in decision-making, supporting the need for accountability in this field (O'Neil, C. (2016). *Weapons of Math Destruction*; Noble, S. U. (2018). *Algorithms of Oppression*). Engaging with these critiques reveals that the intersection of data and policy requires more than just technical expertise—it necessitates a commitment to social justice and the dismantling of systemic inequalities.)

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Chapter 5 Summary : Data’s Mathematical Baptism

Chapter 5 Summary: Data’s Mathematical Baptism

Introduction to Data Science and Its Historical Roots

In this chapter, Chris Wiggins discusses the evolution of data science, focusing on the contributions of notable statisticians and the intersection of data analysis with real-world applications. The chapter highlights key figures such as William Gosset, Ronald Fisher, and Jerzy Neyman, whose methodologies for hypothesis testing and significance formed the backbone of modern statistics.

Gosset and Experimental Design

William Gosset, known for his work at Guinness Brewery, faced challenges with small sample sizes in brewing experiments. To address this, he devised the Student’s t-test,

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enabling the assessment of statistical significance with limited data, prioritizing practical outcomes over strict adherence to universal significance levels.

Fisher and Scientific Inquiry

In contrast, Ronald Fisher viewed statistics as a means to derive scientific knowledge. His work emphasized the need for randomized experiments to eliminate biases, laying the groundwork for hypothesis testing that persists in diverse research today.

Neyman and Decision Theory

Jerzy Neyman shifted the focus from the pursuit of truth to decision-making under uncertainty. He introduced concepts like type I and type II errors and advocated for methodological rigor in hypothesis testing, emphasizing the need for practical applications in research.

The Rise of Bayesian Methods

The chapter also discusses the rise of Bayesian statistics, particularly at Bletchley Park during World War II, where

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cryptography relied on probabilistic reasoning to decode messages. Bayesian approaches allowed for continuous updating of probabilities, establishing a foundation for future data-driven methodologies.

Machine Learning and Data Science Emergence

During the Cold War, the focus on machine learning and artificial intelligence began to emerge. Initial efforts centered around symbolic reasoning but gradually shifted toward data-driven learning methodologies, especially in the context of vast amounts of data generated during the digital age.

The Transformation of Society through Data

Wiggins argues that the integration of mathematical statistics into governance and corporate decision-making fundamentally reshaped societal structures. The normalization of data collection and analysis provided a scientific veneer for the persistence of inequality, reinforcing existing prejudices and promoting discriminatory practices.

Conclusion: The Impact of Data on Society

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The chapter concludes by reflecting on the implications of transforming human attributes into quantifiable data, reinforcing the need for critical scrutiny of how data is interpreted and used. Wiggins posits that understanding this historical context is essential for engaging with contemporary challenges faced in the realm of data science and its ethical implications.

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Chapter 6 Summary : Data at War

Chapter 6: Data at War

Overview

In this chapter, Wiggins explores the evolution and implications of data collection, analysis, and technology in warfare, primarily during World War II and its aftermath. It discusses the development of data-driven techniques, the rise of institutions and theoretical frameworks around data, and the role of powerful organizations in shaping the narrative and application of data.

Bletchley Park and Early Computing

- Bletchley Park emerged as a pivotal site for data analysis during WWII, where mathematicians and engineers, including Alan Turing, developed techniques to break German codes using specialized machines.
- The work at Bletchley combined computation and large-scale data processing, marking a shift toward industrial

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applications of statistics over pure theoretical investigations.

Bayesian Methods and Intelligence

- Turing and his team employed Bayesian methods to refine probabilities of various code-breaking hypotheses, reflecting a practical approach to statistical analysis in a high-stakes environment.
- The efficiency of Bayesian analysis was crucial for rapid decision-making during wartime.

Post-War Innovations

- After the war, the U.S. National Security Agency (NSA) evolved to use vast amounts of data for intelligence gathering, leveraging technologies initially developed for the military.
- NSA significantly influenced the growth of computational

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Chapter 7 Summary : Intelligence without Data

Chapter 7: Intelligence without Data

Overview of the Shift in Data Science

This chapter discusses the transition from traditional views of intelligence and data analysis to a more machine-driven perspective largely dominated by large-scale data collection and machine learning practices. Key contributors to this evolution include scientists like Claude Shannon, Alan Turing, and various influential figures from the fields of machine learning and artificial intelligence (AI).

Historical Context and Early AI Aspirations

- Claude Shannon expressed aspirations for machines that could think, learn, and interact with humans as early as 1952.
- The discourse surrounding AI shifted to focus on computational models rather than on the meaningful analysis

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of enlightening data.
- While early ideas about AI encompassed a wide range of competencies, this broadened vision was soon narrowed, focusing mainly on tasks centered around logic and pre-set rules.

The Influence of Pioneers

- Alan Turing's work emphasized behavior as a key determinant of intelligence, challenging the era's focus on logical thinking.
- Despite contributions from Turing and similar pioneers, the immediate post-war years saw AI research veer towards a symbolic representation rather than data-driven models.

Statistical Developments in the Post-War Era

- The mid-20th century witnessed significant advancements in statistics, driven by practical concerns during World War II.
- Key figures such as Ronald Fisher and Jerzy Neyman brought new perspectives to experimental design and hypothesis testing, contaminating the arena with their debates over the meanings and applications of statistical significance.

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Evolution of Data Practices

- The revolution in machine learning leveraged computational advantages to predict outcomes using correlations rather than causations.
- Initiatives like the Netflix Prize highlighted the pivot toward predictive algorithms fueled by large datasets.

Emergence of Data Science as a Discipline

- In the early 2010s, data science burgeoned as a distinct field, blending statistics, machine learning, and practical software engineering to address real-world problems.
- Critics like Cathy O'Neil asserted that the focus shifted heavily towards prediction capabilities over controlling biases and ensuring fairness.

Societal Implications of Data-driven Algorithms

- The chapter details the implications of data collection practices on various stratifications of social identity (e.g., race, gender), pseudoscientific constructs like intelligence

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and eugenics, and the systemic inequalities they reinforce.
- As predictive technologies proliferated, they also drew critiques regarding their impacts on privacy and democratic governance.

Conclusion: The Path Forward

- The risks associated with totalizing data representations reflect a historical legacy of measurement that requires scrutiny. Solutions prompting more ethical considerations in data analytics are paramount, as the boundary between technological advancement and societal impact narrows.
- Emphasizing contextual understanding of data, the integration of diverse skill sets, and vigilance against reifying statistical results are crucial for future ethical data practice.
In summary, the chapter paints a broad yet intricate picture of how shifts in perspective on intelligence, data collection, and statistical interpretation have led to contemporary challenges within data science, emphasizing the need for more ethical and reflective practices in the face of unprecedented data volumes.

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Chapter 8 Summary : Volume, Variety, and Velocity

Summary of Chapter 8: Volume, Variety, and Velocity

Overview of Big Data Era

In 2012, the term "big data" gained prominence as organizations began harnessing cost-effective methods to manage large volumes of data characterized by high velocity and variety. The chapter reflects on the evolution of data practices, statistical methods, and the societal implications of data collection and analysis.

Historical Context

The chapter introduces R. Blair Smith's encounter with C.R. Smith, president of American Airlines, which led to the development of the SABRE reservation system. This system became an early example of commercial data processing,

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highlighting the increasing interconnection between government and corporate data practices, particularly in the context of surveillance and privacy concerns.

Legislative and Ethical Challenges

The 1977 Privacy Protection Study Commission reported the dangers of unchecked data collection by both the government and corporations, stressing the erosion of individual rights in favor of institutional interests. The challenges of safeguarding privacy amid technological advancements became pressing, with privacy advocates calling for comprehensive data protection laws.

Machine Learning and AI Development

The chapter outlines the shift from traditional artificial intelligence, which sought to emulate human reasoning, to a focus on machine learning and data-driven approaches. It discusses the rise of data science as a field, emphasizing the blending of statistical methods, machine learning, and computational practices to analyze vast data sets.

Emergence of Data Science

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Cathy O'Neil and others highlight that the role of the data scientist evolved from diverse quantitative approaches, with a growing emphasis on practical data analysis, predictive modeling, and the crafting of algorithms. The rise of powerful computational resources allowed for real-time processing of data, transforming industries and institutions.

Ethics and Decision-making

As data analysis capabilities expanded, ethical considerations became paramount. The chapter reflects on the implications of data use in various sectors, including finance and healthcare, where decisions based on flawed or biased data could have catastrophic consequences.

The Future of Data Analysis

The chapter concludes by asserting that data science holds transformative potential, but the blending of data collection, corporate interests, and ethical considerations creates a complex landscape that requires vigilance and advocacy to ensure equitable practices. It emphasizes the need for ongoing discussions about the implications of data-driven decisions in society.

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Chapter 9 Summary : Machines, Learning

Chapter 9: Machines, Learning

Introduction to Machine Learning

In this chapter, the evolution of machine learning is discussed, highlighting its shift from complex reasoning tasks to simpler prediction models. Pat Langley expresses disappointment in how machine learning has narrowed its focus, moving away from reasoning and language understanding.

Historical Context

Langley compares the early ambitions of machine learning to its current state, where the field has become dominated by statistical tools for prediction rather than holistic approaches to intelligence. The initial divergence between numerical methods of pattern recognition and symbolic AI is examined.

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Gosset's Contributions

William Gosset, an engineer at Guinness, developed the t-test to make decisions based on small sample sizes, emphasizing practical applications over purely academic ones. His work focused on maximizing profitability through experimentation in brewing.

Fisher's Significance Testing

Ronald Fisher's work further refined statistical testing methods, introducing the concept of p-values to determine the significance of results. He emphasized the importance of randomization in experimentation to ensure objectivity. Fisher sought to transform statistics into a more rigorous scientific discipline, emphasizing experimental design.

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Chapter 10 Summary : The Science of Data

The Science of Data

Overview

Chapter 10 of "How Data Happened" by Chris Wiggins discusses the evolution and significance of data science against a backdrop of historical and contemporary concerns regarding technology, data collection, and ethical implications. Wiggins asserts that the intersection of data and technology has reshaped power dynamics, societal norms, and people's understanding of intelligence and knowledge.

The Origins of Data Science

- The chapter begins by expressing how technology cannot be categorized as merely good, bad, or neutral. Kranzberg's first law sets the stage for understanding the complex interplay between technology and society.

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- Hanna Wallach's advocacy for fairness in machine learning highlights critical inquiries into the ethical dimensions of algorithms developed for real-world applications, suggesting that the technology community needs to collaborate with social scientists to foster transparent systems.

Historical Context

- The text references key figures such as Quetelet, Nachtingale, Galton, and Fisher, who contributed to statistical methods that intersected with political and social reform.
- Through their work, statistics became a tool to manage societies and populations, answering pressing social questions using data to understand individual behaviors in aggregates.

Evolving Definitions and Practices

- The chapter discusses the varying definitions and debates surrounding data science, particularly how it integrates statistics, machine learning, and computational analysis tailored to real-world challenges.
- Prominent examples of early data applications—such as

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Nielsen ratings and agricultural studies—illustrate how data has been leveraged for economic and social insights.

Algorithmic Governance and Injustices

- Wiggins explores how data analytics, particularly in the realm of policing and social services, often exacerbated existing inequalities rather than alleviating them.
- Key criticisms of data practices, including the work by Safiya Noble and Cathy O’Neil, emphasize how such systems can perpetuate biases against marginalized communities.

Bayesian Methods and Machine Learning

- The chapter goes into the technical details of significant statistical methods, highlighting advances in Bayesian statistics and their importance in real-time scenarios like national defense.
- Turing's influence at Bletchley Park exemplifies how data analysis became integral to military strategies during WWII, showcasing the importance of data within the socio-political sphere.

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Contemporary Data Science

- Wiggins argues that the emergence of data science as a profession encapsulates a blend of skills, including statistical analysis, coding, system management, and communication abilities.
- This evolution reflects broader trends in society towards valuing empirical evidence and data-driven decision-making, seen in sectors like healthcare, finance, and marketing.
- Ethical concerns and discussions about accountability intensify as data science increasingly influences policy, personal liberties, and societal norms.

Conclusion

- The chapter highlights the critical need for robust frameworks and ethical considerations surrounding data collection and analysis.
- Emphasizing data's power to shape societies, Wiggins calls for an awareness of how the choices made in data practices affect public understanding and governance. Ultimately, the ongoing evolution of data science necessitates vigilant scrutiny of its implications and a commitment to using technology for equitable outcomes.

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Chapter 11 Summary : The Battle for Data Ethics

Section	Summary
Introduction to Data Ethics	Growing power of algorithms requires ethical considerations in data usage; scholars and activists are calling for accountability and fairness in algorithmic systems.
The Call for Fairness in AI	At a 2018 symposium, Hanna Wallach emphasized the need for ethical scrutiny and interdisciplinary collaboration in machine learning to address biases and discrimination.
Growing Concerns Over Big Data	Experts like danah boyd and Kate Crawford raised alarms about big data's implications, underscoring issues of privacy and discrimination in technologies like Google search algorithms.
The Role of Intelligence Agencies	Post-9/11 scrutiny of government surveillance highlighted ethical issues in data practices, particularly how they target vulnerable populations.
Corporate Responses to Ethical Challenges	Tech companies have begun hiring AI ethicists, but critics point out that these roles often lack the power to effect real change, revealing a gap between ethics and corporate practices.
The Need for Robust Accountability	Without strong accountability frameworks, algorithm integration could worsen inequality; ethical frameworks alone are insufficient without systemic changes.
Historical Context and Current Implications	Historical data practices show a neglect of ethics for profit and efficiency, highlighting a long-standing tension between social good and inequality.
Conclusion: The Path Forward	Advocacy for reforms prioritizing accountability, transparency, and social justice is essential to ensuring technology promotes equity. Collective action is vital for aligning tech with human values.

Chapter 11: The Battle for Data Ethics

Introduction to Data Ethics

As the power of algorithms and machine learning continues to grow, so does the urgency for ethical considerations in data usage. This chapter discusses the efforts made by a

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growing number of scholars and activists to challenge existing practices, calling for accountability and fairness in algorithmic systems that increasingly shape societal outcomes.

The Call for Fairness in AI

At a 2018 symposium, Hanna Wallach highlighted the need for machine learning researchers to scrutinize their work's ethical implications, especially concerning biases that can perpetuate discrimination. She emphasized the necessity of interdisciplinary collaboration, arguing that understanding social data requires input from social scientists.

Growing Concerns Over Big Data

Previous warnings about big data's implications were echoed by voices like danah boyd and Kate Crawford, who questioned whether the surge in data usage would lead to advancements in technology that benefit society or result in privacy violations and social control. Safiya Noble’s critique of Google search algorithms showcased how biases embedded in technology disproportionately affect marginalized communities, while Cathy O’Neil illuminated

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how unchecked data practices could dehumanize individuals in various sectors.

The Role of Intelligence Agencies

Post-9/11 revelations concerning government surveillance intensified public scrutiny of data practices, raising critical questions about privacy and the ethical dimensions of algorithmic decision-making. Virginia Eubanks presented historical examples demonstrating how such systems often target the most vulnerable populations, leading to broader societal implications.

Corporate Responses to Ethical Challenges

Prominent tech companies began hiring AI ethicists in response to critiques, attempting to engage ethically with their technologies. However, many critics have noted that these roles often lack true power and influence, highlighting a disconnect between ethical ideals and corporate practices.

The Need for Robust Accountability

Critics argue that without substantial frameworks for

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accountability, the integration of algorithms into policymaking and daily life could lead to exacerbated inequality. Ethical frameworks alone cannot guarantee responsible AI; systemic changes are essential to ensure fair use of technology across different sectors.

Historical Context and Current Implications

The historical lineage of data practices demonstrates how ethics have been sidelined in favor of efficiency and profit. Early statistical practices laid the groundwork for contemporary discussions, revealing a persistent tension between the use of data for social good versus the perpetuation of inequality.

Conclusion: The Path Forward

To influence positive change in data ethics, activists and scholars must advocate for reforms that prioritize accountability, transparency, and social justice. Understanding the intricacies of data collection and algorithmic decision-making is vital for promoting a society where technology serves as a tool for equity rather than oppression. The narrative throughout history indicates that

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collective action and informed advocacy are essential for realigning technology with human values.

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Example

Key Point:The urgency for accountability in data ethics is increasingly important in our technology-driven society.

Example:Imagine you're applying for a mortgage online. As you submit your application, you notice a checkbox asking to opt into data collection analytics, which seems harmless but has complex implications. What if the algorithm that assesses your creditworthiness unintentionally factors in biases from historical data, leading to unfair rejection? You realize that the ethical considerations in data use directly impact your life choices and opportunities. This scenario illustrates the urgent need for ethical scrutiny and accountability in AI systems to ensure fairness and prevent discrimination, highlighting the growing importance of advocacy for responsible data practices.

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Critical Thinking

Key Point:The Role of Advocacy in Data Ethics

Critical Interpretation:The chapter asserts that collective action and informed advocacy are crucial in addressing the ethical challenges posed by evolving data practices. However, it's important to recognize that such advocacy may not universally translate into effective policy changes or real-world impact. Critics such as Michael Burawoy argue that activism can be co-opted by those in power, potentially diluting its efficacy and creating a facade of ethical compliance without genuine reform. It invites readers to contemplate the complexities of the struggle for data ethics, suggesting that while advocacy is necessary, it alone may not lead to meaningful outcomes.

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Chapter 12 Summary : Persuasion, Ads, and Venture Capital

Summary of Chapter 12: Persuasion, Ads, and Venture Capital

Introduction to Data and Society

This chapter explores the intersection of data, persuasion, advertising, and venture capital, emphasizing the burgeoning relationship between technology and human behavior. It begins by referencing Herbert Simon's assertion that understanding and designing for an information-rich world is crucial for contemporary governance.

The Impact of Algorithms on Truth

It highlights the dramatic influence algorithms and recommendation systems, particularly on platforms like YouTube, can have on shaping perceptions of reality. The historical backdrop includes Edward L. Bernays' work in

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public relations, which illustrates the potent use of communication strategies in advertising.

The Attention Economy

As the digital landscape advanced, concepts like the "attention economy" emerged, wherein platforms compete for user engagement through tailored advertisements. This commodification of attention is scrutinized, presenting a moral dilemma about what content receives exposure online.

Microtargeting in Political Campaigns

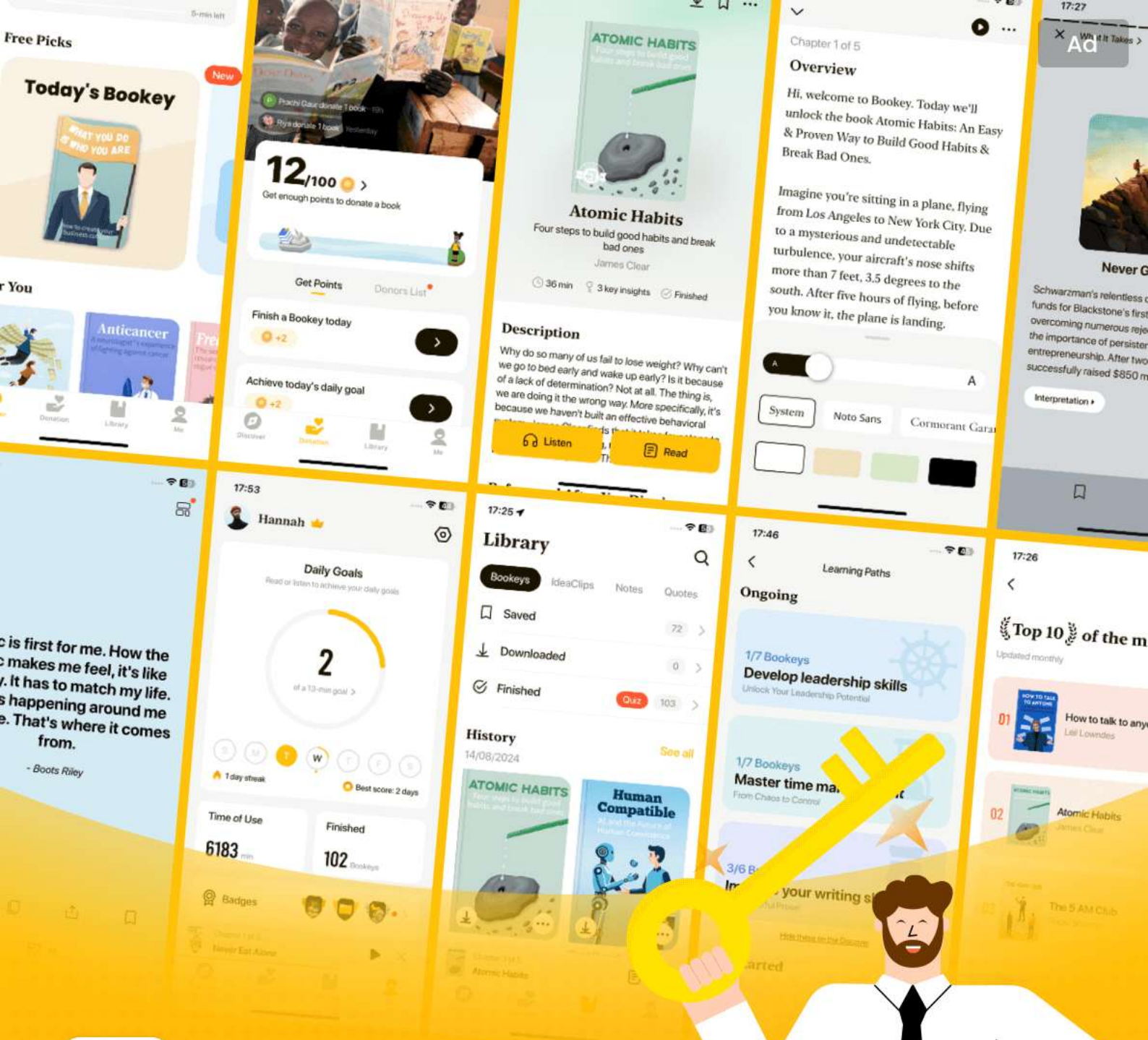
The chapter discusses how political campaigns, notably during the 2016 U.S. Presidential Election, employed sophisticated data analytics—such as those from Cambridge Analytica—to microtarget voters. The deployment of personalized content aimed to manipulate public opinion.

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Chapter 13 Summary : Solutions beyond Solutionism

Solutions beyond Solutionism

Overview of the Content

Chapter 13 discusses the complex relationship between data science, technology, ethics, and societal implications. It highlights the critiques and suggestions made by several influential scholars, including Hanna Wallach, Danah Boyd, and Safiya Noble, who emphasized the importance of fairness and accountability in algorithmic systems. The chapter connects historical perspectives on statistics and data collection to contemporary practices, revealing that many of the challenges posed by today's data-driven world echo past issues.

Key Themes and Arguments

-

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Interdisciplinary Collaboration
: Wallach argues for the inclusion of social scientists in data analytics to understand the biases present in datasets. The need for collaboration between technologists and social scientists is highlighted as essential for developing fair algorithms.

-
Historical Context of Statistics
: The chapter reflects on the evolution of statistics from a purely quantitative tool to one that reflects and influences societal norms. Quetelet’s, Galton’s, and Pearson's contributions to statistical thought are examined, focusing on how statistical methods have been used to reinforce societal structures.

-
Eugenics and Race
: Historical figures like Galton and Pearson contributed to the development of statistical methods that were later used to justify eugenics and racial hierarchies. This carried implications for social policy and public perception of marginalized groups.
-

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Evolving Definitions of Science

: The text discusses the shift in academic perspectives on data collection and analysis, suggesting that the focus has moved from qualitative understanding to a narrow emphasis on quantitative metrics. This transition compromises the depth and breadth of understanding necessary for tackling social issues.

-

Algorithmic Accountability and Ethics

: The chapter critiques the way companies and governments have appropriated ethical concerns regarding algorithms without instituting meaningful accountability measures. Many organizations hired AI ethicists but often stifled their criticisms.

-

Privacy and Data Rights

: Discussions unfold around the implications of privacy rights, especially after the emergence of extensive data collection practices post-9/11. There’s a call for better protections and considerations of individual rights as they relate to large-scale data use.

-

Impact of Data and AI

: The chapter concludes by asserting that as data science

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continues to grow, it must be accompanied by a robust ethical framework that can address the societal implications of this power.

Conclusion

Ultimately, chapter 13 of "How Data Happened" calls for a deeper understanding of data practices and their implications on society, urging for a move away from solutionism towards more nuanced and responsible approaches that consider ethics, fairness, and accountability in the age of big data.

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Critical Thinking

Key Point:Interdisciplinary Collaboration

Critical Interpretation:The author advocates interdisciplinary collaboration, positing that merging social sciences with data analytics can counteract ingrained societal biases. While this perspective is important, it may oversimplify the challenges of integrating disciplines with differing methodologies, potentially overlooking the complexities involved in reconciling quantitative and qualitative insights. Scholars like E. Gibbons (2019) argue that while interdisciplinary approaches are beneficial, they also face significant obstacles, including disciplinary silos and communication gaps.

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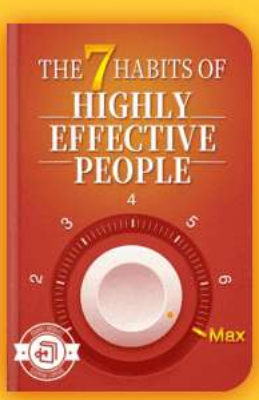
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Chapter 1 | Quotes From Pages 1735-2311

1. Technology is neither good nor bad; nor is it neutral. –Kranzberg’s first law of technology, 1986
2. We teach a course at the University of Michigan called 'the Internet is a trash fire,' and I don’t have to explain to anybody what that means. –Lisa Nakamura, 2019
3. Wallach diagnosed the problem—one outside the traditional disciplinary scope of computer science. Admitting that the solutions to the problem would not come from within computer science, she instead demanded collaboration with those from other fields.
4. New capabilities yielded new powers. These powers threatened to entrench many of the inequalities that so many societies had long struggled, with checkered and uneven success, to undo.
5. The future is already here—it’s just not very evenly

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distributed.” —William Gibson

6. The real danger is the gradual erosion of individual liberties through the automation, integration, and interconnection of many small, separate recordkeeping systems.
7. It is vital not physical, it is biological, it is medical, it is statistical. It needs not less but far more investigation for its solution than any academic physical or biological problem.
8. To understand human society meant understanding these general facts, something made possible by the accumulation of ever-larger amounts of data about that society and its people.
9. The challenges required to scale machine learning in the arenas of commercial, intelligence, and military data promoted the creation of technologies and technologists capable of dealing with ever-larger data sets.
10. In our work, we ought to have robust institutional forms that enable a holding to account, not simply force the production of an account.

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Chapter 2 | Quotes From Pages 2312-2888

1. Technology is neither good nor bad; nor is it neutral." –Kranzberg’s first law of technology, 1986
2. If modern civilisation is distinguished from all other civilisations by its scientific basis, the problems that this civilisation presents must be solved by the methods of Science.
3. The inclination is to replace people with data trails, turning them into more effective shoppers, voters, or workers to optimize some objective.
4. Data is made, not found, and the process of procuring and analyzing it often dramatically loops back to shape the people under official scrutiny.
5. The average captures something existing above and beyond each person, something about the group itself, something about how each individual group member acts.
6. Statistics is in many ways a wealthy man’s game. The poor get cheaper housing, but the same statistics don’t

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necessarily apply to the privileges they claim.

7. Deciding to affirm something scientific, Neyman wrote, ‘doesn’t mean knowing or even believing.’

8. In the words of the historian Alain Desrosières, ‘a political problem’ was translated ‘into an instrument of measurement that allowed arbitration of a controversy.’

9. The art of experimental design and of the valid interpretation of experimental results, in so far as they can be technically perfected, must constitute the core of this claim to the exercise of full intellectual liberty.

10. Collecting data and filling seats: so read the subtitle to a description of American Airlines’ SABRE System that developed in the wake of this conversation.

Chapter 3 | Quotes From Pages 2889-3465

1. Technology is neither good nor bad; nor is it neutral.

2. We put up with this for a long time; we don’t seem to know anything different.

3. The era of Big Data has begun.

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4.Data is made, not found.

5.The power of man over animal life is enormously great. It would seem as though the physical structure of future generations was almost as plastic as clay, under the control of the breeder’s will.

6.To understand poverty, one must seek the cause not in any difference of national qualities, but in external conditions.

7.Data doesn’t simply represent the world. It transforms how we categorize and view the world.

8.Measurement must be regarded as problematic and we can ask not only for the conditions under which events in a person’s life will become a matter of record, but whether it is legitimate for them to become a matter of record.

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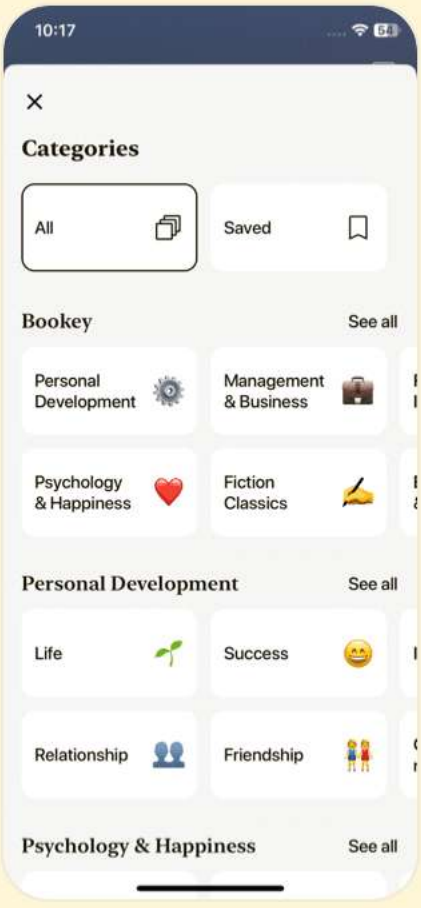
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Chapter 4 | Quotes From Pages 3466-4042

1. Technology is neither good nor bad; nor is it neutral.
2. We put up with this for a long time; we don’t seem to know anything different.
3. As an example: while it’s relatively straightforward to obtain and analyze data from Twitter users, these data are hardly representative of, say, the US population overall.
4. Science must lay down afresh the laws of life.
5. Statistics is the work of many hands.
6. The future is already here—it’s just not very evenly distributed.
7. Our world involves just such a rearrangement of power.
8. We must have robust institutional forms that enable a holding to account.
9. Data is made, not found.
10. The general conclusion emphasized by nearly every investigator is that, as regards 'intelligence', the Germanic stock has on the average a marked advantage over the

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South European.

Chapter 5 | Quotes From Pages 4043-4619

1. Technology is neither good nor bad; nor is it neutral.
2. We put up with this for a long time; we don’t seem to know anything different.
3. Few computer scientists or engineers would consider developing models or tools for analyzing astronomy data without involving astronomers. So, why, then, are so many methods for analyzing social data developed without the involvement of social scientists?
4. Raw data is an oxymoron... all data collection comes through human choice about what to collect, how to classify, who to include and to exclude; all collection involves cognitive biases and radically different infrastructures for categorizing, storing, and processing that information.
5. The real danger is the gradual erosion of individual liberties through the automation, integration, and

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interconnection of many small, separate recordkeeping systems.

Chapter 6 | Quotes From Pages 5197-5773

1. Technology is neither good nor bad; nor is it neutral.
2. The inclination is to replace people with data trails, turning them into more effective shoppers, voters, or workers to optimize some objective.
3. In the age of the internet, journal publication may seem ridiculously slow—but it was a radical transformation in the public availability and circulation of data.
4. A good computer program will not only compute the desired answers, but will also present them in a meaningful way.
5. Statistics does not simply represent the world. It transforms how we categorize and view the world.
6. Data is made, not found, and the process of procuring and analyzing it often dramatically loops back to shape the people under official scrutiny.

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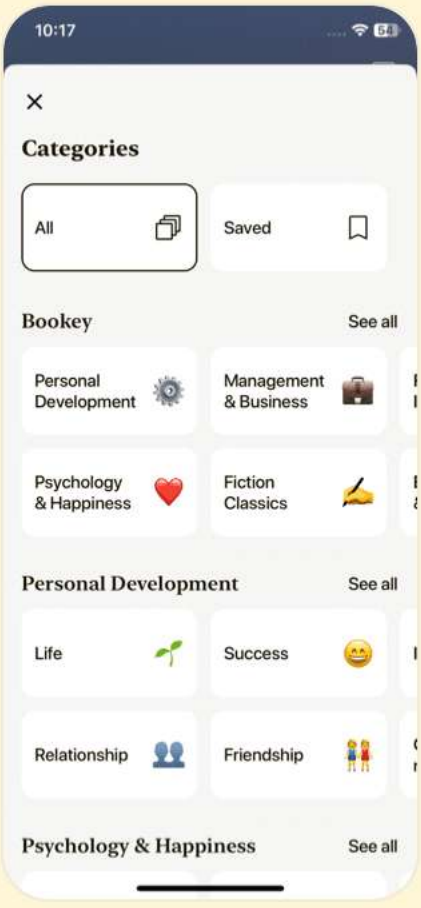
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Chapter 7 | Quotes From Pages 5774-6350

1. Technology is neither good nor bad; nor is it neutral.
2. We need to start thinking outside the algorithmic boxes typically embraced by the machine learning community and instead focus on the opportunities, challenges, and implications involved in developing and using machine learning methods to analyze real-world data about society.
3. The future is already here—it’s just not very evenly distributed.
4. Society includes within itself the germs of all the crimes committed, and at the same time the necessary facilities for their development.
5. Data is made not found, whether in 1600 or 1780 or 2022.
6. The general conclusion emphasized by nearly every investigator is that, as regards 'intelligence,' the Germanic stock has on the average a marked advantage over the South European.

Chapter 8 | Quotes From Pages 6351-6927

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1. Technology is neither good nor bad; nor is it neutral.
2. The inclination is to replace people with data trails, turning them into more effective shoppers, voters, or workers to optimize some objective.
3. Data must be viewed not just as numbers, but as reflections of our collective realities.
4. Statistical laws can prevail for a mass even when the constituent individuals are too numerous or too inscrutable for their actions to be understood in any detail.
5. Exploring the new world of data is not just about understanding it, but about shaping our future with that understanding.
6. Privacy must be weighed today against the value gained from the collection and availability of information at central points or data banks.
7. Fears of the power of algorithms to drive computational politics at the scale of entire nation-states...
8. Data is made, not found, and the process of procuring and

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analyzing it often dramatically loops back to shape the people under official scrutiny.

Chapter 9 | Quotes From Pages 6928-7504

1. Technology is neither good nor bad; nor is it neutral.
2. We teach a course at the University of Michigan called ‘the Internet is a trash fire,’ and I don’t have to explain to anybody what that means.
3. If modern civilization is distinguished from all other civilizations by its scientific basis, the problems that this civilization presents must be solved by the methods of Science.
4. The inclination is to replace people with data trails, turning them into more effective shoppers, voters, or workers to optimize some objective.
5. To understand human society meant understanding these general facts, something made possible by the accumulation of ever-larger amounts of data about that society and its people.

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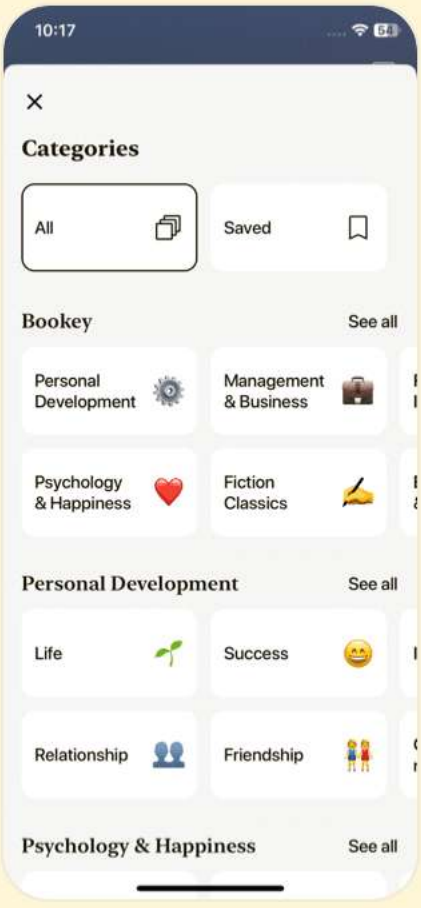
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Chapter 10 | Quotes From Pages 7505-8081

1. Technology is neither good nor bad; nor is it neutral.
2. We must start thinking outside the algorithmic boxes typically embraced by the machine learning community and instead focus on the opportunities, challenges, and implications involved in developing and using machine learning methods to analyze real-world data about society.
3. What we do not learn from history, we are doomed to repeat.
4. Statistical methods alone cannot fix the deep structural issues that data and algorithms serve to reinforce. We need to critically engage with the socio-political context surrounding data use.
5. Data is made, not found, and the process of procuring and analyzing it often dramatically loops back to shape the people under official scrutiny.

Chapter 11 | Quotes From Pages 8659-9235

1. Technology is neither good nor bad; nor is it

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neutral.

2. The inclination is to replace people with data trails, turning them into more effective shoppers, voters, or workers to optimize some objective.
3. Data is made, not found; and often the processes of procuring and analyzing it dramatically loop back to shape the people under official scrutiny.
4. Numerical accountability, with its promises of transparency and rule-bound objectivity, typically gained prominence in conditions of distrust about experts.
5. If modern civilization is distinguished from all other civilizations by its scientific basis, the problems that this civilization presents must be solved by the methods of Science.

Chapter 12 | Quotes From Pages 9236-9812

1. Technology is neither good nor bad; nor is it neutral.
2. If we find an immense difference between the mind of some race and that of European race, we must seek the

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cause not in any difference of national qualities, but in external conditions.

3. The inclination is to replace people with data trails, turning them into more effective shoppers, voters, or workers to optimize some objective.
4. The great biologist of the future will be like the great physicist of today, a mathematician trained and bred.
5. Every social problem belongs to a class embracing the hardest of all problems—it is vital not physical, it is biological, it is medical, it is statistical.
6. It is this slippage from the process to the thing that made Yule’s conclusion so ticklish to interpret.

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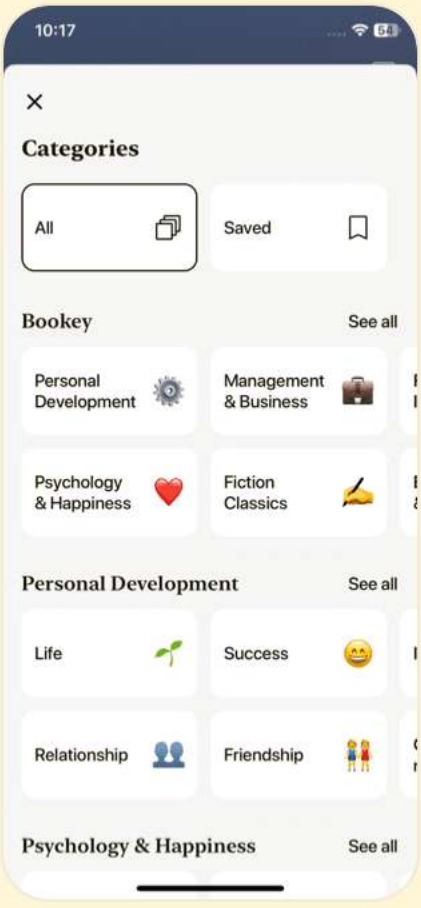
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Chapter 13 | Quotes From Pages 9813-10389

1. This principle is advantageous to the partisans of a revolution... Abrupt movements are never made without a certain loss of live force.
2. The power of man over animal life is enormously great... It would seem as though the physical structure of future generations was almost as plastic as clay, under the control of the breeder’s will.
3. If modern civilization is distinguished from all other civilizations by its scientific basis, the problems that this civilization presents must be solved by the methods of Science.
4. The fundamental objection to this form of statistical reasoning is that some of the most important influences at work are of a kind which cannot be measured quantitatively.
5. The 'new information technologies' seem to have given birth to a new social virus—'data-mania.'
6. There is still a certain chance of error depending on the

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number of factors correlated both with pauperism and with proportion of out-relief which have been omitted, but obviously this chance of error will be much smaller than before.

7. We can never hope to lay bare all the truth we must have, if we are to cope successfully with the problems growing out of the heterogeneous commingling of races which our defective immigration laws are forcing upon us.
8. Our future lies in the hands of people whose demands must be met through fundamentally relevant and fully informed statistical reasoning. Without that, we risk building systems that fail to serve those most in need.
9. The average captures something existing above and beyond each person, something about the group itself, something about how each individual group member acts.
10. Statistics doesn’t simply represent the world. It transforms how we categorize and view the world.

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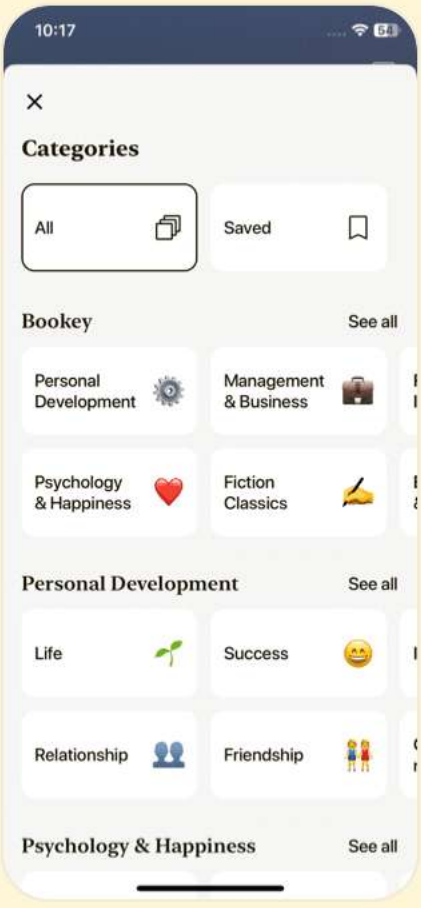
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How Data Happened Questions

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Chapter 1 | The Stakes| Q&A

1.Question

What are the implications of Hanna Wallach's call for fairness, accountability, and transparency in machine learning?

Answer:Hanna Wallach's call for fairness, accountability, and transparency in machine learning emphasizes the need for computer scientists and engineers to engage with social scientists and community stakeholders when developing algorithms. Without this collaboration, algorithms may perpetuate existing biases and inequalities. Her proposal reflects a growing recognition that machine learning should not only be about technical proficiency but also about ethical considerations and social impact.

2.Question

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How do historical examples of data misuse inform current data practices?

Answer:Historical examples, such as Frederick Hoffman's racial statistical analysis that justified discrimination against Black Americans, highlight the dangers of misusing data to reinforce existing prejudices. These lessons inform current practices by underscoring the importance of ethical data collection, the need for critical scrutiny of algorithmic outcomes, and the recognition that data is not neutral but shaped by political and social contexts.

3.Question

Why is the distinction between correlation and causation important in the context of data science and predictive modeling?

Answer:The distinction between correlation and causation is crucial because it prevents misinterpretation of data. Correlation indicates a relationship between two variables but does not imply that one causes the other. Failing to recognize this can lead to incorrect conclusions and policies

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based on misconceptions about how variables interact, which can have significant real-world consequences, especially in areas like healthcare, economics, and social policy.

4.Question

What can we learn from the historical context of statistical thinking in relation to data ethics today?

Answer:Understanding the historical context of statistical thinking reveals that data practices have often been used to justify social hierarchies and discrimination. This highlights the need for contemporary data ethics frameworks to address power dynamics and ensure inclusivity. Today's practitioners must be aware of the historical potential for data to harm, advocating for equity and justice in data collection and analysis.

5.Question

In what ways does the role of the data scientist differ from traditional statisticians?

Answer:The role of the data scientist often encompasses practical data wrangling, software engineering, and business

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acumen, in addition to statistical analysis. While traditional statisticians may focus primarily on theoretical concepts and model validation, data scientists are tasked with applying statistical methods to real-world data at scale, often prioritizing predictive accuracy over theoretical rigor.

6.Question

How has the perception of data changed from the past to the present in society?

Answer:Data has shifted from being seen primarily as administrative tools for governance to becoming central components of business strategy, social engineering, and personal surveillance. The contemporary view recognizes data's potential for both empowerment and exploitation, leading to increased scrutiny over privacy, consent, and algorithmic accountability.

7.Question

What is the significance of Kranzberg’s first law of technology in understanding the societal impact of data?

Answer:Kranzberg’s first law—"Technology is neither good

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nor bad; nor is it neutral"—is significant in understanding that technology reflects the values and intentions of those who design and implement it. This perspective encourages critical examination of how data technologies are deployed, reminding us that their societal impact can vary widely based on how they are used and who is in control.

8.Question

What challenges do emerging data-driven technologies pose to ethical standards and accountability?

Answer:Emerging data-driven technologies often operate with limited transparency, making it difficult to hold developers and organizations accountable for harmful outcomes. The rapid pace of innovation can outstrip ethical guidelines and regulatory frameworks, leading to issues like algorithmic bias and privacy violations, which necessitates a continual evolution of ethical standards in tandem with technological advancement.

9.Question

How does the concept of 'data feminism' contribute to current discussions on data ethics?

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Answer:'Data feminism' contributes to discussions on data ethics by advocating for practices that acknowledge and address the power imbalances embedded in data collection and analysis processes. It emphasizes inclusive methodologies that amplify marginalized voices, ensuring that data practices promote social justice rather than reinforce existing inequalities.

Chapter 2 | Social Physics and l’homme moyen| Q&A

1.Question

How did Adolphe Quetelet influence modern statistics?

Answer:Quetelet, a Belgian astronomer, developed the concept of 'social physics' and introduced the idea of the 'average man'. He inspired the use of statistical methods to understand societal phenomena, promoting the notion that large-scale social data could reveal general laws governing human behavior, similar to how observations in astronomy could predict celestial movements.

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2.Question

What critique did Florence Nightingale make about the legislation of her time?

Answer:Nightingale criticized the hasty changes to laws without adequate data collection or analysis of results. She pointed out that legislative changes were often made without understanding their impacts, advocating for a more evidence-based approach to governance.

3.Question

Why is the concept of 'vulgar statistics' significant in the critique of statistical methods?

Answer:'Vulgar statistics' refers to the shallow or simplistic application of statistics to understand complex social issues, as criticized by skeptics who believed that mere numbers could not capture the qualitative realities of human life. This concept underscores the need for careful interpretation and contextual understanding of statistics rather than treating them as absolute truths.

4.Question

In what way did Francis Galton contribute to eugenics

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and the understanding of intelligence?

Answer:Galton pioneered the field of eugenics, arguing that intelligence and other traits could be inherited. He established statistical methods for measuring these traits and influenced public policy and social attitudes toward race and intelligence, promoting the notion of 'general intelligence' (g) and the hereditary nature of these characteristics.

5.Question

How did Karl Pearson’s work separate itself from earlier approaches to statistics?

Answer:Karl Pearson transitioned statistics from descriptive accounts of populations to a more rigorous mathematical framework. He introduced concepts like correlation and regression, emphasizing the importance of quantifying relationships between variables, thus laying the foundation for modern statistical theory.

6.Question

What role did data labor play in the development of AI and machine learning?

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Answer:Data labor is essential in preparing and cleaning datasets, which significantly influences the effectiveness of AI and machine learning. Though often invisible, the efforts of countless individuals to organize and label data enable the operation of algorithms designed to analyze and learn from these datasets.

7.Question

How did the war effort during World War II shape modern statistical practices?

Answer:World War II required rapid data processing and analysis for military operations, leading to innovations in statistical methods and the development of computing machines. The success of wartime statistics established a model for applying statistical reasoning to real-world problems, influencing various fields such as medical research, economics, and data science.

8.Question

What impact did the development of the Internet and big data have on statistics and analysis?

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Answer:The Internet and big data revolutionized statistics by producing vast amounts of data that could be analyzed, leading to the emergence of fields like 'data science'. This shift emphasized prediction and optimization over traditional statistical validation methods, thus changing the nature of how we understand data-driven decision-making.

9.Question

How have societal perceptions of data privacy evolved with the advent of big data?

Answer:Societal perceptions of data privacy have shifted from a focus on individual rights against government intrusion to a broader concern encompassing corporate surveillance and data commodification. This change highlights the complexities of privacy in the digital age, where personal information is often collected and used without explicit consent.

10.Question

What is the significance of the term 'data scientist' in contemporary discussions about data analysis?

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Answer:The term 'data scientist' signifies a merging of data analysis with skills from statistics, machine learning, and software engineering. It reflects the evolving nature of work in analyzing large datasets and emphasizes the need for multidisciplinary expertise in harnessing data for decision-making and innovation.

Chapter 3 | The Statistics of the Deviant| Q&A

1.Question

What is the key message regarding the intersection of technology, data, and society in Chapter 3?

Answer:The key message is that while technology and data analytics have vast potential to drive social progress and improve decision-making, they also carry significant risks of reinforcing existing inequalities and bias. The chapter emphasizes the need for critical engagement with these technologies to ensure they are used ethically and do not exacerbate societal issues.

2.Question

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How did Adolphe Quetelet's ideas influence the understanding of social physics and statistics?

Answer:Quetelet introduced the concept of 'l'homme moyen' or the 'average man,' arguing that studying statistics could yield insights into societal behavior and norms. He emphasized the importance of understanding populations through numerical analysis rather than merely through individual anecdotes, effectively laying the groundwork for modern statistical methods.

3.Question

What are some historical consequences of applying statistical analysis to social issues, as discussed in this chapter?

Answer:The application of statistical analysis to social issues often led to the reinforcement of hierarchical structures, as seen in eugenics. Figures like Francis Galton and Karl Pearson used statistical data to promote concepts of racial superiority, which justified discriminatory practices and policies, adversely affecting marginalized communities.

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4.Question

In what ways does the chapter highlight the challenges of achieving fairness and accountability in algorithmic decision-making?

Answer:The chapter notes that biases in data often mirror societal inequalities, which can lead to flawed algorithms that perpetuate discrimination. It emphasizes the importance of interdisciplinary collaboration to address these issues and advocates for transparency and accountability in the development and implementation of algorithms.

5.Question

What lessons can be drawn from the historical analysis of data and statistics presented in this chapter?

Answer:Historical analysis reveals that data and statistical practices are not neutral but shaped by social, political, and economic contexts. The chapter underscores the necessity of approaching data critically, being aware of the underlying structures of power that influence data collection and interpretation.

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Chapter 4 | Data, Intelligence, and Policy| Q&A

1.Question

How does the concept of 'average man' relate to the creation of statistics?

Answer:The 'average man', introduced by Quetelet, represents a new way of understanding societal norms through statistical averages. Quetelet argued that by analyzing large populations and their traits, one could derive meaningful insights about societal standards and behaviors, effectively elevating the concept of average to something more real and influential in shaping societal policies.

2.Question

What role did data play in the evolution of eugenics and racial theories?

Answer:Data was central in the development of eugenics, as figures like Galton and Pearson used statistical methods to argue for innate differences among races. By creating and analyzing data on population traits, they sought to justify

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discriminatory practices and social hierarchies, ultimately leading to policies that targeted certain racial groups.

3.Question

What are the ethical implications of data analysis as discussed in the chapter?

Answer:The ethical implications are profound, as data analysis can reinforce existing inequalities and biases if not carefully scrutinized. The use of data can lead to unjust policies based on flawed assumptions, thus emphasizing the need for critical engagement with the data and understanding the social contexts in which it is applied.

4.Question

What is the significance of the correlation versus causation debate in statistics?

Answer:The correlation versus causation debate highlights the risk of misinterpretation in statistical analysis. Correlation can suggest a link between two variables without proving that one causes the other, which can lead to misguided conclusions and policies if not addressed properly.

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5.Question

How has the role of data scientists evolved in today's data-driven world?

Answer:Data scientists have evolved from merely analyzing data to implementing complex machine learning algorithms designed not only for analysis but for making predictions based on vast data sets. Their role now encompasses a mix of analytical, technical, and domain-specific skills, reflecting the diverse nature of data interactions in modern commerce and governance.

6.Question

What lessons can we draw from the historical misuse of statistics in justifying discrimination?

Answer:The historical misuse of statistics teaches us the importance of ethical standards in data collection and analysis, as well as the need for transparency in methods. Moreover, it emphasizes the necessity of including diverse perspectives in research to avoid reinforcing systemic inequalities.

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7.Question

How did the insights from statistics influence public policy historically?

Answer:Insights from statistics have historically influenced public policy by providing empirical evidence to support or challenge existing policies. For instance, data demonstrating the distribution of poverty helped formulate welfare policies, showcasing both the potential benefits and pitfalls of relying on statistical analysis in governance.

8.Question

What was the impact of the introduction of statistical methods on the perception of racial differences?

Answer:The introduction of statistical methods lent a veneer of objectivity to discussions about racial differences, leading to a scientific approach that often ignored broader social and historical contexts. This resulted in entrenched stereotypes and legitimized social divisions based on flawed interpretations of data.

9.Question

What does the chapter imply about the relation between

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technology and societal values?

Answer:The chapter implies that technology, including statistical methods and data analysis, is not inherently neutral; it reflects and reinforces societal values and power dynamics. Thus, the development and application of technology must be critically examined to ensure it advances social justice and equity.

10.Question

How did early data collection methods contribute to contemporary issues of privacy and surveillance?

Answer:Early data collection methods laid the groundwork for the massive accumulation of personal data that we see today. The normalization of data collection for administrative purposes, without robust protections, has led to privacy concerns and an erosion of rights, particularly for marginalized communities.

Chapter 5 | Data’s Mathematical Baptism| Q&A

1.Question

What does Chris Wiggins suggest about the relationship

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between data and power in his writing?

Answer:Chris Wiggins highlights the intricate dynamics between data, algorithms, and societal structures. He suggests that while data has the potential to enhance fairness and accountability, it often amplifies existing inequalities and biases, particularly when wielded by those in power. The ability to collect, process, and analyze large datasets can lead to systemic discrimination if not checked by ethical considerations and inclusive practices.

2.Question

How did the rise of big data and machine learning alter traditional views on privacy and individual rights, according to the chapter?

Answer:The rise of big data shifted the perspective on privacy rights from a collective concern to an individualistic view. While privacy advocates in the 1970s recognized that data collection disproportionately affected marginalized groups, the focus of later discussions increasingly narrowed

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to personal freedoms, undermining broader calls for accountability and protection against systemic abuses of data.

3.Question

What was the significance of figures like Joan Fisher Box and Ronald Aylmer Fisher in the development of statistical methods?

Answer:Figures like Joan Fisher Box and Ronald Aylmer Fisher were critical in refining statistical methods for practical applications, such as hypothesis testing and experimental design. Their work laid the framework for modern inferential statistics, emphasizing the importance of statistical significance in empirical research across various fields, including agriculture and medicine.

4.Question

Describe the impact of Quetelet's idea of the 'average man' on societal understanding of human differences.

Answer:Quetelet's concept of the 'average man' introduced the idea that statistical averages could represent societal norms and standards, effectively reifying these averages as objective truths about human behavior and physical

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characteristics. This facilitated the categorization of individuals and groups, influencing policies around race, intelligence, and social justice, often leading to discriminatory practices.

5.Question

How does Wiggins connect data practices with historical movements and societal changes over time?

Answer:Wiggins connects data practices to historical movements such as eugenics and social reform, illustrating how statistical methods were used to both understand and control populations. He demonstrates the evolution of data collection, from the use of data for governance in the 19th century to its manipulation in the modern age, showing that data has always been a tool for both empowerment and oppression.

6.Question

What role do algorithmic systems play in reinforcing existing social inequalities?

Answer:Algorithmic systems can perpetuate and even

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exacerbate existing social inequalities by relying on biased data that reflects historical injustices. When algorithms are developed without consideration of ethical implications or representative data, they risk embedding systemic biases into decision-making processes, affecting areas such as criminal justice, healthcare, and employment.

7.Question

Why is the discussion of data ethics becoming increasingly important in the context of artificial intelligence?

Answer:The discussion of data ethics is becoming crucial as artificial intelligence systems are deployed to make decisions that impact individuals and society at large. With the potential for algorithms to cause harm through biased outputs and lack of accountability, ethical considerations must guide the development and application of AI to ensure fairness, transparency, and respect for individual rights.

8.Question

What examples do you observe of challenges faced by women in the early history of data analysis and statistics?

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Answer:Women played pivotal roles in the early history of data analysis and statistics, often working in the shadows of male counterparts. Figures like Florence Nightingale and Alice Lee contributed significantly to data collection and early statistical methods, yet their contributions were frequently overshadowed by their male peers, revealing the gender biases that have persisted in the field.

9.Question

In what way has personal data collection evolved, and how does it affect individual privacy?

Answer:Personal data collection has evolved from isolated, manual record-keeping practices to expansive, automated systems that aggregate vast amounts of personal information. This evolution significantly impacts individual privacy by creating detailed profiles that can be accessed and analyzed without the individual's knowledge or consent, often leading to surveillance and discriminatory practices.

10.Question

What overarching theme about data and society can be derived from Wiggins' analysis?

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Answer:An overarching theme in Wiggins' analysis is the duality of data as a tool for empowerment and control. While data has the power to inform positive change and enhance understanding of societal issues, it also poses risks of reinforcing unequal power dynamics, exacerbating biases, and infringing on individual freedoms when not governed by ethical standards.

Chapter 6 | Data at War| Q&A

1.Question

What does Kranzberg's first law of technology imply about the role of human agency in shaping technology?

Answer:Kranzberg's first law highlights that technology is shaped by human choices and societal contexts, meaning it is influenced by values, ethics, and the specific situations we face, thus framing technology as neither inherently good, bad, nor neutral.

2.Question

How does Hanna Wallach emphasize the need for interdisciplinary collaboration in data analysis?

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Answer:Wallach stresses that computer scientists should collaborate with social scientists when analyzing data about society, as they bring essential perspectives and expertise that are necessary to understand the implications of algorithmic systems.

3.Question

In what ways did Edward Snowden's disclosures affect public perception and discourse around privacy and data collection?

Answer:Snowden's disclosures revealed the extent of government surveillance, which reignited discussions about privacy rights, the ethics of data collection, and the balance of power between citizens and the state.

4.Question

What are some examples of biases that can emerge from algorithms according to Safiya Noble and Cathy O’Neil?

Answer:Noble points out biases in Google search outcomes that render certain groups, especially Black women, invisible, while O’Neil discusses how data-driven algorithms can reinforce existing inequalities in sectors like education

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and criminal justice.

5.Question

How does W. E. B. Du Bois challenge Frederick Hoffman's interpretation of racial differences?

Answer:Du Bois critiques Hoffman by arguing that racial disparities in outcomes are not due to innate differences but rather the result of socio-economic factors and historical injustices that continue to shape opportunities.

6.Question

How did the advances in statistical methods during WWII change the application of data science?

Answer:The advances made during WWII, particularly in codebreaking, led to the establishment of large-scale data analysis capabilities that emphasized the importance of real-time processing and statistical methods in military operations, which later influenced civilian applications.

7.Question

Why is it essential to differentiate between correlation and causation when analyzing data?

Answer:Distinguishing between correlation and causation is

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crucial to avoid misinterpretations that could lead to false conclusions. Misunderstanding this distinction can result in policies based on incorrect assumptions about the relationships among variables.

8.Question

What role did women play at Bletchley Park despite being largely unrecognized in historical narratives?

Answer:Women at Bletchley Park performed critical roles as operators of early computers and in data processing, yet their contributions have often been overshadowed or erased in accounts that focus predominantly on their male colleagues.

9.Question

What transformations did data undergo from being a base for scientific inquiry to being a powerful tool for surveillance and control?

Answer:Data transitioned from being a means to understand patterns in society to a powerful tool for profiling individuals, where collected information can be used for both beneficial analysis and, more troublingly, for surveillance, discrimination, and social control.

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10.Question

How did the emergence of data science redefine the understanding of intelligence and decision-making?

Answer:Data science shifted the focus from purely logical reasoning and understanding to include predictive analytics based on data, allowing organizations to make decisions based on patterns rather than just causal theories or intuitive knowledge.

11.Question

What is the significance of the Netflix Prize competition in the evolution of data science?

Answer:The Netflix Prize competition exemplified how public competitions for improvements in predictive accuracy could spur innovation in machine learning and data science, leading to the development of collaborative and ensemble modeling techniques.

12.Question

In what ways did the relationship between data, privacy, and power shift in the late 20th century?

Answer:The late 20th century saw a shift where concerns

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about privacy were increasingly focused on government surveillance while neglecting corporate power and data collection practices, leading to a less critical public discourse around personal data.

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Chapter 7 | Intelligence without Data| Q&A

1.Question

What does it mean to have “Intelligence without Data”?

Answer:The phrase points to the notion that intelligence should not solely rely on data-driven systems or metrics to understand phenomena. It underscores the idea that human judgment, reasoning, and creativity should complement analytical and computational techniques. Intelligence can manifest in contexts where quantitative data is scarce or even non-existent, emphasizing the importance of qualitative insights and experiential knowledge.

2.Question

How can data science impact societal values?

Answer:Data science has the potential to reflect and reinforce societal values, but it can also challenge existing norms. When data is used to inform policy or business decisions, the metrics and goals prioritized inherently shape what is

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considered important or valuable in society. This means that data-driven decisions can lead to significant social changes, such as increased efficiency in services, but may also perpetuate biases if the data reflects historical injustices.

3.Question

What role do biases play in algorithms and data analysis?

Answer:Biases can infiltrate algorithms and data analysis processes in various ways, including the selection of data sets, the framing of questions, and the interpretation of results. If the underlying data used to train models is biased or unrepresentative, the algorithms will perpetuate and amplify these biases in their outputs, leading to unfair or discriminatory outcomes. The key is to recognize and address these biases to promote fairness and accountability.

4.Question

What are the ethical implications of surveillance in data practices?

Answer:Surveillance in data practices raises serious ethical questions regarding privacy, consent, and the equitable

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treatment of individuals. As data collection proliferates, often without individuals' awareness, the potential for misuse or exploitation of personal information becomes a profound concern. Ethical frameworks must be established to govern data usage and ensure that the rights and interests of individuals, particularly marginalized groups, are respected.

5.Question

How has the focus on algorithms shaped our understanding of intelligence?

Answer:The focus on algorithms has often narrowed our understanding of intelligence to mere data processing and predictive accuracy, sidelining more complex human capacities such as emotional intelligence, creativity, and critical thinking. This trend can yield powerful applications in technology but risks creating a view of intelligence that is reductive and ignores the richness of human experience and cognition.

6.Question

Why is it important not to view statistics as purely mathematical?

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Answer:Statistics should not be viewed as merely mathematical because it is fundamentally tied to human experience and social realities. The act of collecting and interpreting data involves subjective choices and cultural contexts which shape the understanding of phenomena. A purely mathematical perspective can lead to misleading conclusions and fails to acknowledge the social implications and ethical considerations inherent in statistical inquiry.

7.Question

What lessons can we draw about the historical development of data practices?

Answer:The historical development of data practices teaches us that the ways we collect, interpret, and apply data are deeply intertwined with power dynamics and societal values. As data practices have evolved, they have often reflected the priorities of those in power, necessitating critical examination and advocacy to ensure that data serves the broader interests of society rather than perpetuating inequality and injustice.

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8.Question

How does the concept of the 'average man' reflect societal views on individuals?

Answer:The concept of the 'average man' reflects a societal tendency to generalize and categorize people based on statistical measures, often leading to the reification of social norms and expectations. This can marginalize those who deviate from the average, reinforcing stereotypes and systemic inequalities. The method of defining individuals through averages can obscure the diversity and complexity of human experiences and identities.

9.Question

What critical perspectives are necessary in the age of big data?

Answer:In the age of big data, critical perspectives that challenge assumptions, interrogate the ethics of data collection, and examine the social implications of data usage are essential. These perspectives should address issues such as privacy, bias, accountability, and the potential for data to

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both empower and oppress different groups. A multifaceted approach, incorporating insights from social sciences, ethics, and technology, will enable a more nuanced understanding of data's role in society.

10.Question

What proactive steps can be taken to address the ethical implications of data use?

Answer:Proactive steps include establishing clear regulations regarding data privacy, implementing transparency measures that allow individuals to understand how their data is being used, promoting ethical frameworks in data science education, encouraging businesses and governments to adopt responsible data practices, and fostering dialogue among stakeholders, including technologists, ethicists, policymakers, and affected communities.

Chapter 8 | Volume, Variety, and Velocity| Q&A

1.Question

How can we effectively address the potential biases in machine learning algorithms?

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Answer:To effectively address potential biases in machine learning algorithms, it is essential to incorporate fairness, accountability, and transparency into the design and deployment processes. Collaborating with social scientists and ethicists during the development phase can help identify possible biases inherent in the data used for training algorithms. Regular audits and assessments of algorithms in real-world applications should be conducted to monitor outcomes and mitigate biases. Moreover, implementing diverse data sets that accurately represent the affected populations ensures that algorithms do not perpetuate existing inequalities.

2.Question

What role did the development of big data play in reshaping business practices?

Answer:The development of big data fundamentally reshaped business practices by enabling companies to collect,

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store, and analyze vast amounts of customer and transactional data at unprecedented scales. This capability facilitated personalized marketing strategies through targeted advertising, improved customer engagement, and operational efficiencies. As organizations began to recognize the immense potential of data for informing decisions, they adopted analytics to drive performance metrics, enhance product offerings, and tailor services to meet specific customer needs, ultimately leading to more data-driven decision-making processes.

3.Question

What are the ethical implications of data collection and analysis in today's digital world?

Answer:The ethical implications of data collection and analysis in today's digital world include significant concerns about privacy, consent, and surveillance. Individuals often unknowingly surrender vast amounts of personal information, raising questions about the transparency of data practices and the extent of informed consent. Moreover, the

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potential for misuse of data—leading to discrimination or targeting vulnerable populations—poses serious risks. Ethical frameworks need to be established to ensure that data is collected responsibly and that its usage aligns with principles of fairness and respect for individual rights.

4.Question

What motivated the shift from traditional AI approaches to data-driven methods in machine learning?

Answer:The shift from traditional AI approaches to data-driven methods in machine learning was motivated by the recognition that vast amounts of data could provide better predictions than rigid rules based on limited human expertise. With increasing computational power and the availability of large data sets, researchers found that machine learning algorithms could adapt to complex patterns and improve their performance without the constraints of symbolic AI. This focus on predictive accuracy over traditional rule-based reasoning allowed organizations to leverage data to solve real-world problems effectively.

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5.Question

How does the concept of 'algorithmic transparency' relate to public trust in technological systems?

Answer:Algorithmic transparency relates to public trust in technological systems by ensuring that the processes and criteria used by algorithms are accessible and understandable to users. When individuals are aware of how algorithms function, how their data is used, and the rationale behind automated decisions, they are more likely to trust those systems. Conversely, a lack of transparency can lead to skepticism and concerns over bias or misuse, decreasing public confidence in technology. Building transparency mechanisms—such as explainable AI—into systems is crucial for fostering trust and accountability.

6.Question

What is the significance of the term 'data science' in the contemporary landscape of technology and research?

Answer:The term 'data science' signifies a merging of statistical analysis, machine learning, and practical

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computing skills tailored to extract insights from large datasets. In the contemporary landscape, data science represents an interdisciplinary approach that combines technology, domain knowledge, and critical thinking to solve complex problems. As organizations increasingly rely on data-driven decision-making, data science has become critical for industries ranging from healthcare to finance, emphasizing the importance of both analytical skills and ethical considerations in handling data.

Chapter 9 | Machines, Learning| Q&A

1.Question

What motivates the call for collaboration between computer scientists and social scientists regarding machine learning?

Answer:Hanna Wallach's call for collaboration stems from the recognition that algorithms and technologies developed within computer science often lack social context, leading to biased outcomes when applied to social data. She emphasizes the

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necessity of involving social scientists to ensure fairness, accountability, and transparency in the models developed for analyzing real-world data.

2.Question

How does the discussion on the potential for harm from algorithms connect to historical contexts of data collection and analysis?

Answer:The discussion highlights how the exploitation of algorithms often mirrors historical patterns of systemic inequality, suggesting that while algorithms can offer insights, they also have the potential to reinforce biases and exacerbate existing social injustices. This echoing of historical injustices is central to critiquing modern uses of data, linking it to the legacy of misuse seen in practices like eugenics and discriminatory policies.

3.Question

What role does the concept of the 'average man' play in the development of statistical methods?

Answer:The concept of the 'average man', introduced by Adolphe Quetelet, serves as a foundation for modern

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statistics, influencing how data about populations is interpreted and portrayed. It provides a framework that transforms average values into perceived social norms, often leading to misunderstandings of individual behaviors when generalized.

4.Question

How do advances in machine learning today reflect or compound historical issues related to data and bias?

Answer:Advances in machine learning often replicate historical issues of bias, as datasets used to train these models can reflect systemic inequalities found in society. Machine learning's reliance on large datasets can perpetuate these biases if not critically analyzed, similar to how historical eugenic practices misused data to enforce discriminatory ideologies.

5.Question

What warning does Virginia Eubanks give about data collection's implications for marginalized communities?

Answer:Eubanks warns that the technological innovations in

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data analysis, while promising, often disadvantage low-rights environments, reinforcing surveillance and punitive measures against marginalized communities. This underlines the need for scrutiny in how such technologies are designed and implemented.

6.Question

How does the discussion surrounding ethical AI relate to the broader societal implications of machine learning?

Answer:The ethical AI discussion emphasizes accountability and fairness in algorithmic decision-making processes, suggesting that without robust ethical frameworks, machine learning technologies can reinforce and exacerbate inequities, impacting vulnerable populations more acutely.

7.Question

What does the term 'data science' encompass in the context of its evolution in industry and academia?

Answer:Data science has evolved from a vague concept to a defined field that integrates statistics, machine learning, and data engineering to solve practical problems, particularly in

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business contexts. It reflects a growing need for professionals who can navigate the complexities of large datasets in various applications.

8.Question

How have historical fears surrounding data privacy resurfaced in contemporary discussions about big data and surveillance?

Answer:Historical fears of data privacy have resurfaced in contemporary discussions as individuals become increasingly concerned about how their personal data is collected, shared, and used by corporations and governments, echoing past anxieties about surveillance and control, particularly in light of events such as the Cambridge Analytica scandal.

9.Question

What can we learn from the historical misuse of data in shaping modern policies and technologies?

Answer:Understanding the historical misuse of data underscores the necessity of establishing regulatory frameworks and ethical guidelines in the development and application of modern technologies, aiming to prevent the

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repetition of past injustices and ensuring that technologies serve the public good rather than exacerbate inequalities.

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Chapter 10 | The Science of Data| Q&A

1.Question

What is the significance of the quote 'Technology is neither good nor bad; nor is it neutral.' in the context of the ethics of data science?

Answer:This quote highlights that technology, including data science, inherently carries values and implications that can affect society positively or negatively. The ethical considerations of how data is collected and used must be central to any advancements in technology, reminding us that decisions in data practices carry moral weight.

2.Question

How does Wallach's admonition to consider fairness and accountability relate to the historical misuse of data and algorithms?

Answer:Wallach's call for collaboration between technologists and social scientists echoes historical critiques where data and algorithms were used to reinforce biases, such as in the cases of racial profiling and discrimination. By

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ensuring diverse perspectives in data development, we can work to prevent the perpetuation of historical injustices.

3.Question

In what ways did statistical modeling shift after WWII as highlighted in the chapter?

Answer:After WWII, the focus of statistical modeling shifted from understanding human behavior through descriptive statistics to making predictions through complex models. It became less about causal explanations and more about optimizing algorithms for predictive accuracy, leading to the rise of machine learning.”},{

Chapter 11 | The Battle for Data Ethics| Q&A

1.Question

What does the phrase 'Technology is neither good nor bad; nor is it neutral' suggest about the impact of technology on society?

Answer:It suggests that technology has inherent biases and implications that can shape societal values and structures, rather than being an objective tool that exists outside of these contexts.

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2.Question

What was Hanna Wallach's main message during her talk at the conference in Montréal?

Answer:Wallach called for a revolution in the machine learning community to incorporate fairness, accountability, and transparency in the development and use of algorithms, signaling a need for interdisciplinary collaboration with social scientists.

3.Question

How did the scandals surrounding AI ethics, like the dismissal of Timnit Gebru from Google, reflect broader issues in the tech industry?

Answer:They highlighted the tension between critical ethical inquiries and corporate interests, revealing how companies often co-opt ethical discussions for public relations while failing to implement meaningful changes.

4.Question

Why did data-driven approaches in AI and machine learning face criticism after the 2016 US Presidential election?

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Answer:These approaches were criticized for enabling disinformation campaigns and microtargeting that could manipulate and polarize the electorate, raising ethical concerns about the use of data in political campaigns.

5.Question

What are the potential dangers of algorithmic systems as discussed in the context of surveillance and data collection?

Answer:Algorithmic systems can reinforce existing inequalities by making discriminatory decisions, as they operate on biased data, and they often involve opaque processes that hinder accountability and transparency.

6.Question

In what ways did the historical context of eugenics shape the development of statistics, according to the text?

Answer:Eugenics influenced the categorization of individuals and groups based on perceived innate characteristics, leading to policies that were justified using statistical methods, often at the expense of marginalized populations.

7.Question

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What is meant by 'data is made, not found,' in the context of data collection and analysis?

Answer:This phrase emphasizes that data collection involves deliberate choices about what to measure and how to classify individuals, which can shape the resulting insights and conclusions.

8.Question

How has the notion of 'the average man' been problematic in statistical discourse?

Answer:The concept of the 'average man' can obscure individual differences and reinforce stereotypes, while also being used to justify social hierarchies and discrimination based on claimed innate characteristics.

9.Question

What role do human workers play in the functioning of AI systems, despite the emphasis on automation?

Answer:Human workers often perform crucial 'data janitorial work,' including cleaning and classifying data, which makes AI systems function effectively; their labor, however, is

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frequently invisible.

10.Question

How did the juxtaposition of academic statistics with practical applications evolve throughout the 20th century?

Answer:Academic statistics gradually moved towards rigorous mathematical formulations, often distancing itself from real-world data, while practical applications in industry emphasized the need for flexible, data-driven approaches.

Chapter 12 | Persuasion, Ads, and Venture Capital| Q&A

1.Question

What are the main ethical implications of data collection and analysis discussed in Chapter 12?

Answer:The ethical implications center around how the tools of data analysis can reinforce existing inequalities, the dangers of algorithmic bias, and the need for accountability in data-driven decision making. Researchers emphasize that data technologies often perpetuate patterns of oppression,

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particularly against marginalized communities.

2.Question

How did the concepts of fairness and accountability in data become prominent in discussions about technology?

Answer:Prominent voices like Hanna Wallach highlighted the urgency of integrating interdisciplinary insights into algorithmic systems, advocating for collaboration with social scientists to address biases and ensure that technologies serve justice rather than harm.

3.Question

What is the main concern regarding the relationship between technology companies and their critics, as exemplified by figures like Timnit Gebru and Margaret Mitchell?

Answer:The main concern is that while technology companies hire AI ethicists to address ethical issues, these hires often face silencing or restrictions when their critiques become too pronounced or challenge the company's practices.

4.Question

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What are the contrasting values of Fisher and Neyman in the context of hypothesis testing?

Answer:Fisher valued the pursuit of truth and the development of scientific knowledge through rigorous inquiry, while Neyman emphasized decision-making and balancing error rates instead of focusing solely on truth.

5.Question

In the context of data ethics, what role does the concept of 'reification' play?

Answer:Reification refers to treating abstract concepts as real entities, which can lead to misinterpretations, such as treating statistical averages as definitive truths about individuals or groups, thereby erasing the complexity of human lives.

6.Question

How did Quetelet’s ideas about averages and the 'average man' contribute to the development of statistical thinking in society?

Answer:Quetelet's concept of the 'average man' transformed how societies approach data collection and social phenomena, suggesting that numerical averages could define

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social norms and inform policies for better governance, albeit often at the expense of recognizing individual differences.

7.Question

What historical events influenced data practices as discussed in Chapter 12?

Answer:Historical events including the civil rights movements, the rise of big data in conjunction with technological advancements, and crises such as World War II and political repression shaped the contexts in which data practices developed an ethical dimension.

8.Question

How do issues of surveillance and privacy play into the conversations around data ethics?

Answer:Surveillance technologies, particularly post-9/11, raised concerns about privacy invasions, data collection by governments and corporations without consent, and the need for rights around personal data to prevent misuse and discrimination.

9.Question

What is the 'Attention Economy' and how is it relevant to

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the discussions in Chapter 12?

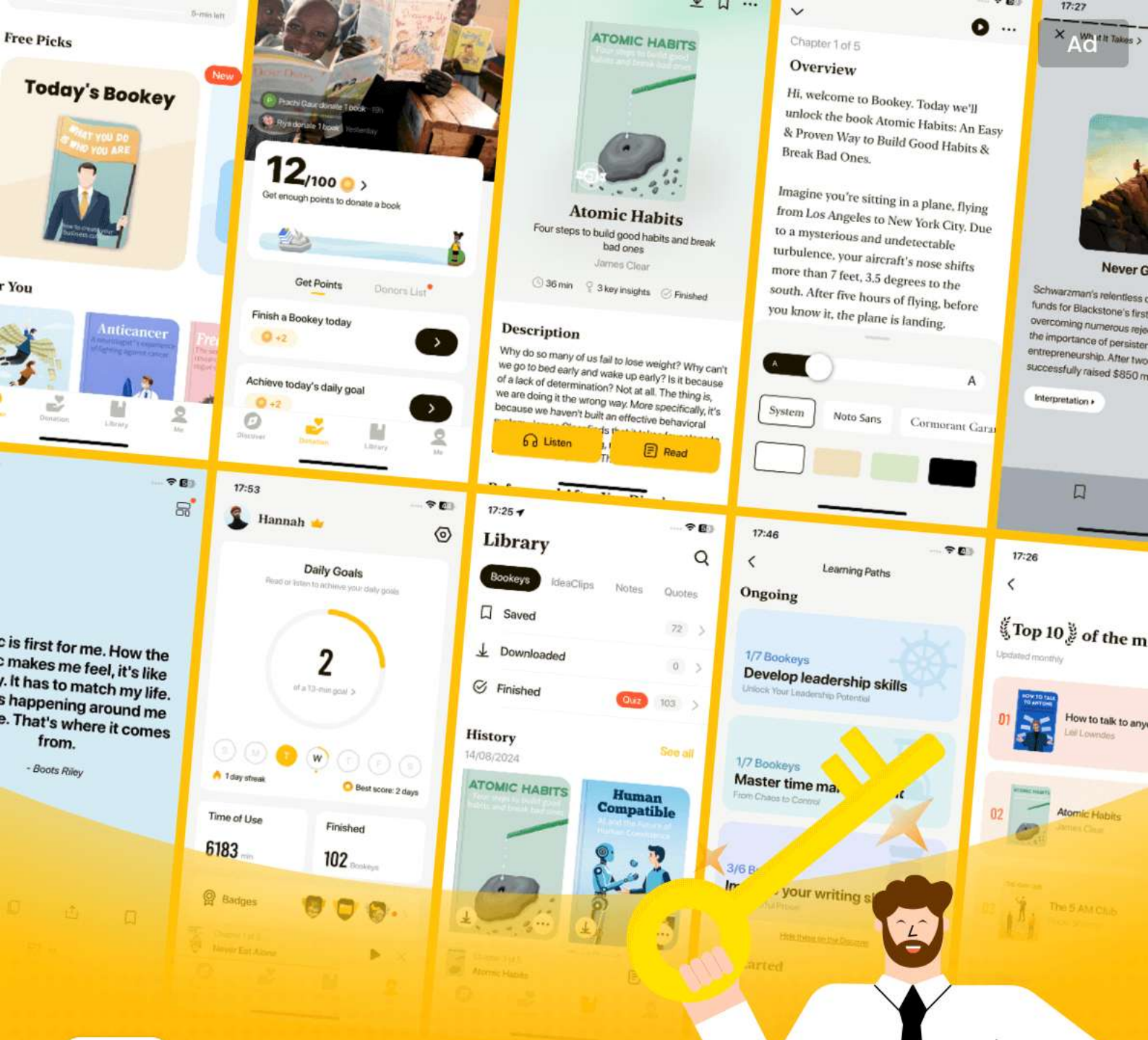
Answer:The 'Attention Economy' refers to the value placed on capturing and maintaining user attention through digital platforms, which has ethical implications regarding manipulation, privacy, and the commodification of personal data in online spaces.

10.Question

How did the definition and role of 'data scientists' evolve in the corporate world according to Chapter 12?

Answer:The role of 'data scientists' evolved to encompass a blend of statistical analysis, machine learning, and engineering skills focused on practical applications, particularly in optimizing business outcomes, sometimes sidelining ethical considerations in pursuit of performance metrics.

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Chapter 13 | Solutions beyond Solutionism| Q&A

1.Question

What is the central message of Wallach's call for collaboration in machine learning and social sciences?

Answer:Wallach emphasizes the necessity for computer scientists to collaborate with social scientists when developing algorithms to ensure fairness and accountability, as many algorithms impact society in significant ways. This highlights the idea that data analysis should not be performed in isolation but rather in a context that incorporates social implications and values.

2.Question

How do Quetelet's ideas about the 'average man' relate to modern data practices?

Answer:Quetelet's concept of the 'average man' proposed that societal patterns can be understood through averages derived from data. In modern practices, this notion manifests through statistical analyses that categorize and 'average' individuals,

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often reinforcing stereotypes and systemic inequalities.

3.Question

What historical context does the author provide regarding the evolution of statistical thought and its implications?

Answer:The author traces statistical thought from Quetelet's social physics to Galton's eugenics, demonstrating how data collection and analysis have both empowered and oppressed marginalized communities throughout history. This evolution reveals an ongoing struggle over the power dynamics embedded within data practices.

4.Question

In what ways does the book highlight the ethical implications of data science?

Answer:The book underscores the ethical concerns arising from the collection and interpretation of data, such as issues of bias, discrimination, and surveillance. It advocates for a critical examination of who benefits from data science and how it can reproduce existing inequalities rather than dismantle them.

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5.Question

What role did Bletchley Park play in the development of modern statistics and data analysis?

Answer:Bletchley Park served as a pivotal site for the application of statistics to wartime codebreaking, leading to innovations in data processing and analysis techniques. It established a legacy that influenced later developments in computational statistics, especially in how large volumes of data could be processed and understood.

6.Question

Why is it significant that privacy concerns evolved alongside technological advancements in data collection?

Answer:The evolution of privacy concerns reflects the growing awareness of the potential harms associated with data collection and surveillance in the context of technological capabilities. As data collection became more pervasive, so too did discussions about the ethical use of personal data and the need for regulations to protect individual rights.

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7.Question

What does the author suggest regarding the future of data ethics and regulation?

Answer:The author advocates for stronger regulations and ethical frameworks governing data use, emphasizing the importance of accountability and transparency in algorithms. The call for systemic change aligns with a broader recognition that data science must prioritize equitable outcomes and protect marginalized communities.

8.Question

How has the perception of statistics changed from Quetelet's time to the present?

Answer:Statistics have transitioned from a tool for understanding social phenomena in a qualitative manner to an approach that often emphasizes quantitative metrics and numerical supremacy, which can obscure human experiences and perpetuate inequalities.

9.Question

What does the phrase 'data is made, not found' imply about the nature of data collection?

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Answer:This phrase suggests that data does not simply exist in a vacuum but is produced through specific processes, choices, and definitions set by researchers and institutions. Understanding this can illuminate biases and assumptions embedded in data and its analysis.

10.Question

How do the actions of individuals in the history of data collection shape our contemporary understanding of data science?

Answer:The historical actions of individuals, particularly in governance and social reform movements, underscored the significance of data-driven decision-making. These legacies influence contemporary data practices and provide critical lessons on the ethical considerations involved in data science today.

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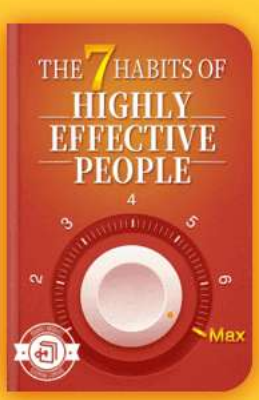
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How Data Happened Quiz and Test

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Chapter 1 | The Stakes| Quiz and Test

1. Data's role in society is purely beneficial according to Chapter 1 of 'How Data Happened'.
2. Kranzberg's first law highlights the importance of critically analyzing technology's implications.
3. Algorithmic decisions have no impact on marginalized communities according to the discussions in Chapter 1.

Chapter 2 | Social Physics and l’homme moyen| Quiz and Test

1. Adolphe Quetelet, a Belgian astronomer, established the study of social physics and emphasized the importance of data in understanding societal phenomena.
2. Quetelet's concept of l’homme moyen was about individual uniqueness rather than statistical averages.
3. Governments began to exploit statistical models in the early 20th century to justify social policies, particularly

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related to race and class.

Chapter 3 | The Statistics of the Deviant| Quiz and Test

1. Quetelet's concept of the 'average man' aimed to apply statistical methods to understand societal behavior and advocated for abrupt societal changes.
2. Francis Galton's belief in eugenics suggested that intelligence and other attributes could be inherited, leading to the promotion of selective breeding for societal improvement.
3. Mahalanobis's work on statistics contributed to reinforcing colonial data collection methods instead of reorienting them to serve independent India's needs.

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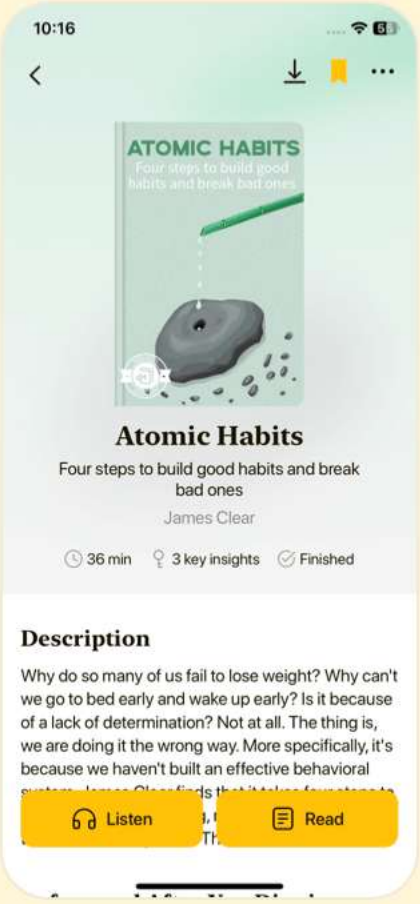
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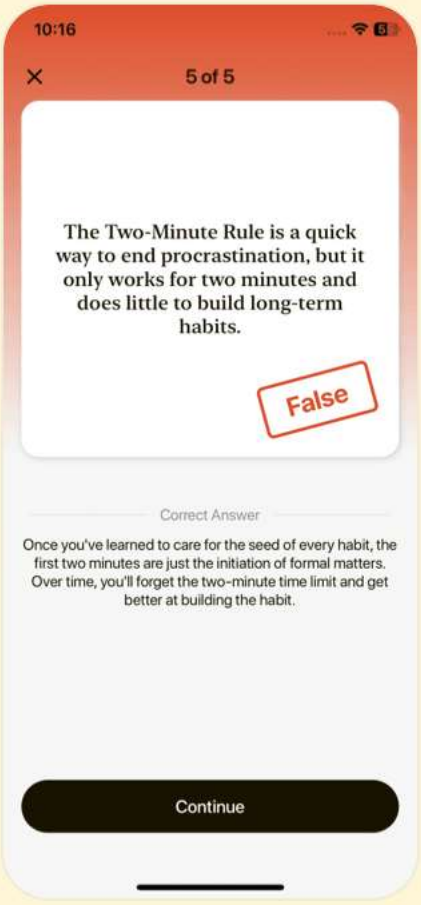
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Chapter 4 | Data, Intelligence, and Policy| Quiz and Test

1. Chapter 4 discusses how data collection can reinforce existing social inequalities.
2. Adolphe Quetelet's work focused on promoting the use of casual relationships to understand societal issues.
3. The chapter emphasizes that data and algorithms are neutral tools that do not require ethical considerations.

Chapter 5 | Data's Mathematical Baptism| Quiz and Test

1. William Gosset developed the Student's t-test to assess statistical significance with small sample sizes.
2. Ronald Fisher believed that statistical methods should prioritize practical outcomes over scientific knowledge.
3. Bayesian statistics were primarily used at Bletchley Park for probabilistic reasoning during World War II.

Chapter 6 | Data at War| Quiz and Test

1. Bletchley Park was significant for its role in data analysis during World War II where

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mathematicians created methods to break codes.

2. The U.S. National Security Agency (NSA) remained unaffected by the advancements in data processing technologies after World War II.

3. The chapter suggests that ethical dilemmas arise when algorithms are used for decision-making processes in big data contexts.

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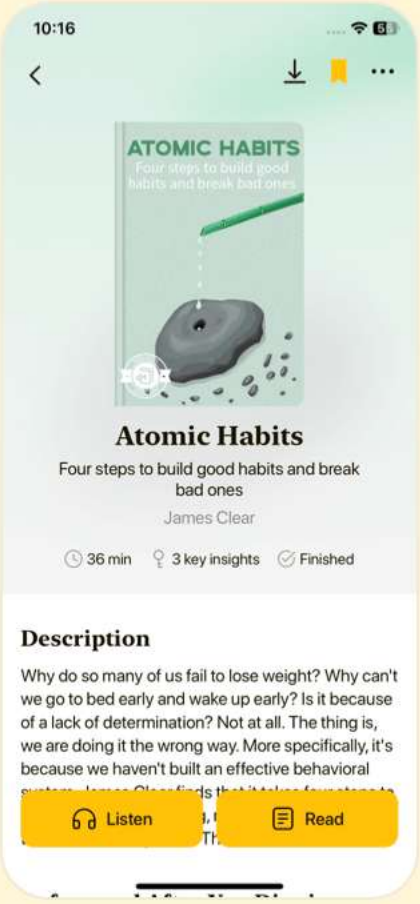
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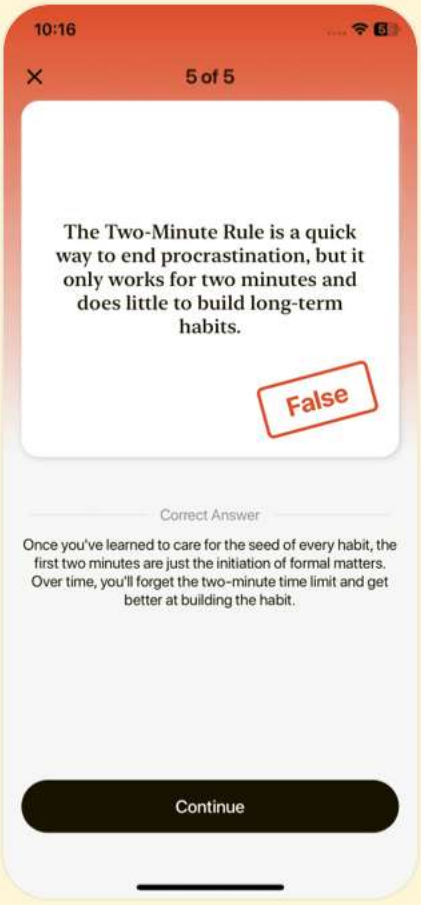
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Chapter 7 | Intelligence without Data| Quiz and Test

1. Claude Shannon expressed aspirations for machines that could think, learn, and interact with humans as early as 1952.
2. Alan Turing's work emphasized logical thinking as the primary determinant of intelligence.
3. The early 2010s saw data science emerge as a distinct field that blends statistics, machine learning, and practical software engineering.

Chapter 8 | Volume, Variety, and Velocity| Quiz and Test

1. The term 'big data' became prominent around the year 2012 as organizations began managing large volumes of data.
2. The SABRE reservation system was developed by American Airlines in collaboration with the government to enhance surveillance measures.
3. Ethical considerations in data analysis have become less important as data analysis capabilities have expanded.

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Chapter 9 | Machines, Learning| Quiz and Test

1. Machine learning has shifted its focus from complex reasoning tasks to simpler prediction models.
2. William Gosset focused solely on academic applications in his development of statistical methods.
3. Bayesian methods are criticized for their objectivity and reliance on randomization in experimentation.

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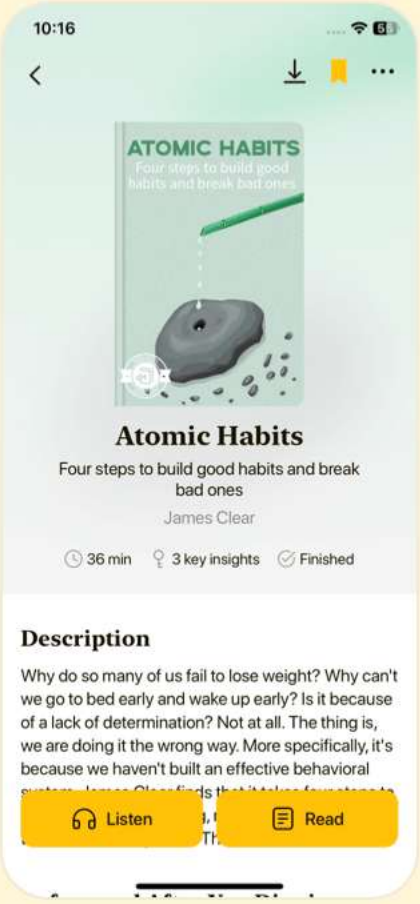
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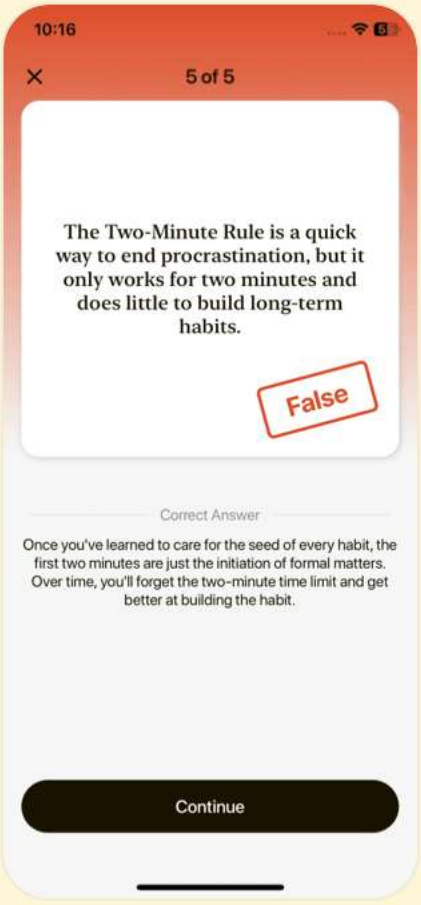
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Chapter 10 | The Science of Data| Quiz and Test

1. The chapter claims that technology can be distinctly categorized as good, bad, or neutral.
2. Wiggins emphasizes the importance of ethical considerations in the development of algorithms for real-world applications.
3. Bayesian statistics were not particularly significant in the domain of national defense during WWII according to the chapter.

Chapter 11 | The Battle for Data Ethics| Quiz and Test

1. The rise of algorithms and machine learning has made ethical considerations in data usage increasingly less important.
2. Virginia Eubanks presented historical examples demonstrating that surveillance systems often protect privileged populations rather than target vulnerable ones.
3. Tech companies hiring AI ethicists has successfully bridged the gap between ethical ideals and corporate

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practices.

Chapter 12 | Persuasion, Ads, and Venture Capital| Quiz and Test

1. Herbert Simon believed that understanding an information-rich world is essential for contemporary governance.
2. The attention economy refers to the competition among platforms to decrease user engagement.
3. Venture capital has no impact on ethical considerations in technology companies that use data-driven applications.

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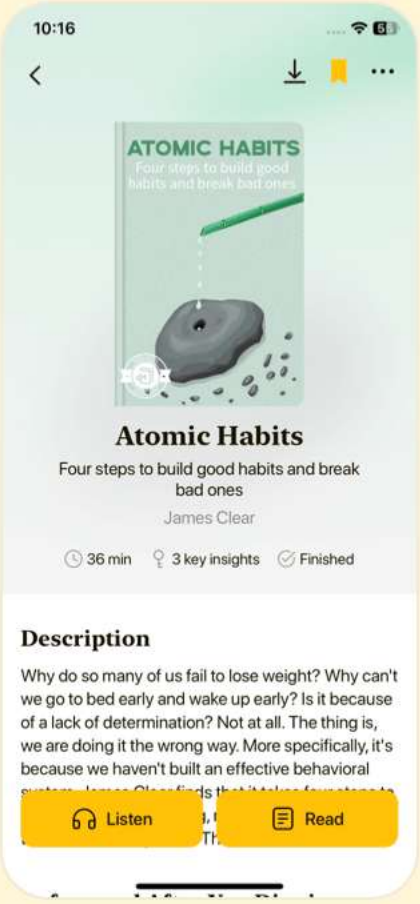
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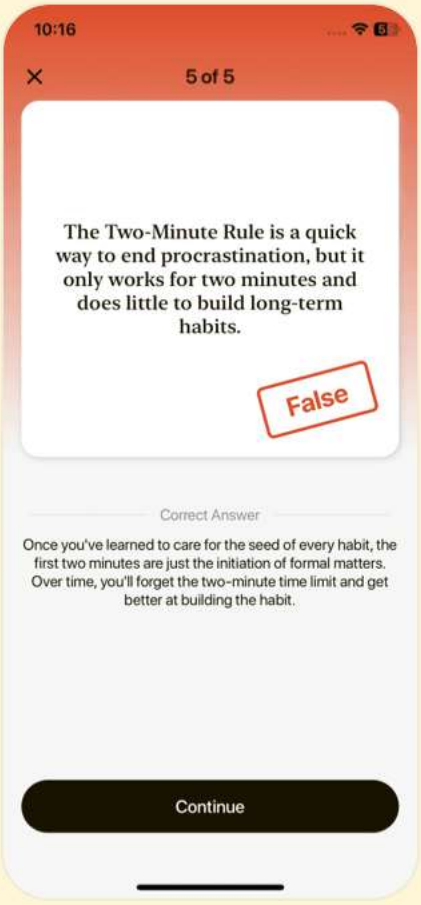
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Chapter 13 | Solutions beyond Solutionism| Quiz and Test

1. Chapter 13 emphasizes the importance of interdisciplinary collaboration between technologists and social scientists in data analytics.
2. The historical context of statistics is irrelevant to understanding contemporary data practices.
3. Algorithmic accountability and ethics are adequately addressed by organizations without the need for further scrutiny.

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